A Deep Learning and Explainable AI Framework for Tuberculosis Diagnosis Using Chest X-ray Images

MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

HARSHITHA BANDLAMUDI-228W1A1270

RADHA VENKATA TARUN MOTUMARRI-238W5A1212

Under the Guidance of

Dr.G.Kalyani

Associate Professor

DEPARTMENT OF INFORMATION TECHNOLOGY

V R SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)

Approved by AICTE &Accredited by NBA

KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

(Affiliated to JNTUK: Kakinada, Approved by AICTE, Autonomous)
(An ISO certified and NBA accredited institution)
Kanuru, Vijayawada – 520007

CERTIFICATE

This is to certify that this project report titled **"A DEEP LEARNING AND EXPLAINABLE AI FRAMEWORK FOR TUBERCULOSIS DIAGNOSIS USING CHEST X-RAY IMAGES"** is a bonafide record of work done by **HARSHITHA BANDLAMUDI(228W1A1270) AND RADHA VENKATA TARUN MOTUMARRI (238W5A1212)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

(Dr.G.Kalyani)	**(Dr. M. Suneetha)**
Associate Professor	Professor &Head
Dept. of Information Technology	Dept. of Information Technology

EXTERNAL EXAMINER SIGNATURE

Date of examination:

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R.SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S.No	Item	Description
1	Project Title	A Deep Learning and Explainable AI Framework for Tuberculosis Diagnosis Using Chest X-ray Images
2	Student Names & Numbers	Harshitha Bandlamudi(228W1A1270) Radha Venkata Tarun(238W5A1212)
3	Name of The Guide	Dr.G.kalyani
4	Research Group	Artificial Intelligence in Medical Imaging
5	Application Area	Health care
6	Aim of the Project	To develop a deep learning-based system for accurate and automated detection of tuberculosis from chest X-ray images.
7	Project Outcomes	• Accurate AI system for detecting tuberculosis from chest X-rays • Explainable and deployable model for real-world clinical use

Student Signatures

1.
2.

Signature of the Guide

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for giving me this opportunity for fulfilling my project. I am grateful to our Principal **Dr. A.V.RATNA PRASAD**, for his encouragement and support all through the way of my project.

On the submission of this Project report, I would like to extend my honour to **Dr. M.Suneetha**, Head of the Department, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my project guide **Dr.G.Kalyani, Associate Professor** for **her** guidance and assistance in completing this project successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficulty on my part to finish the project work.

TABLE OF CONTENTS

ACKNOWLEDGEMENT	iv
LIST OF FIGURES	vii
LIST OF TABLES	viii
LIST OF EUATIONS	ix
ABSTRACT	x
CHAPTER-1 Introduction	1-6
1.1Origion of the Problem	1
1.2 Basic definitions and Background	1
1.3Problem Statement with Objectives and Outcomes	3
1.4 Related Applications	5
CHAPTER-2 Review of Literature	7-18
2.1Description of Existing Systems	7
2.2Summary of Literature Study	10
2.3Software Requirement Specification	12
CHAPTER-3 Proposed Method	19-36
3.1Design Methodology	19
3.2System Architecture Diagram	21
3.3Description of Algorithms	24
3.4 Description of datasets, Requirements and Tools	33
CHAPTER-4 Results and Observations	39-62
4.1Stepwise description of Results	39
4.2Test case results/Result Analysis	59
4.3Observations from the work	60

CHAPTER-5 Conclusion and Future work 63-65

5.1 Conclusion 63

5.2 Future study 64

References

Appendix

LIST OF FIGURES

Figure	Page
Figure3.1: Architecture Diagram	22
Figure3.2: Sample Images in the dataset	34
Figure 4.1: Accuracy &Loss graphs for Convnext	41
Figure 4.2: Confusion Matrix of Convnext	42
Figure 4.3: ROC of Convnext	43
Figure 4.4: Accuracy graph for Vit	44
Figure 4.5: Loss graph for Vit	44
Figure 4.6: Confusion Matrix for Vit	45
Figure 4.7: ROC curve for Vit	46
Figure 4.8: Accuracy &Loss graphs for Ensemble	47
Figure 4.9: Confusion Matrix for Ensemble	47
Figure 4.10: ROC curve for Ensemble	48
Figure 4.11: Accuracy graph for Swin	49
Figure 4.12: Loss graph for Swin	50
Figure 4.13: Confusion Matrix for Swin	50
Figure 4.14: ROC curve for Swin	51
Figure 4.15: Captum for Swin	52
Figure 4.16: Captum for Convext	52
Figure 4.17: Captum for Vit	53
Figure 4.18: Captum for Ensemble	54
Figure 4.19: Gradcam++ for TB and Normal case	54
Figure 4.20: Occlusion for TB and Normal case	55
Figure 4.21: User Interface of TB detection web application	56
Figure 4.22: Prediction results display of Web Application	57

LIST OF TABLES

Table Name	Page
Table 2.1: Literature Review	10
Table 4.1:Result analysis for models	60

LIST OF EQUATIONS

Equation	Brief Description	Page
Eq - (1)	Precision	40
Eq - (2)	Accuracy	40
Eq - (3)	Recall	40
Eq - (4)	F1-score	40
Eq - (5)	ROC_AUC	41

ABSTRACT

The objective of this project is to design an automatic tuberculosis detector using deep learning models through chest X-ray images to help health care professionals diagnose Tuberculosis. For designing this model, the Tuberculosis Chest X-ray dataset provided by Kaggle was employed, which consists of 4,200 images (both healthy and infected). Data augmentation, class-weighting loss, and early stopping approaches were employed in addressing class imbalance in the dataset. ConvNeXt, Vision Transformer, Swin Transformer, and ensemble models including the ConvNeXt and Vision Transformers are four models trained for detecting Tuberculosis. According to the test accuracy results, ConvNeXt obtained the best performance with an accuracy of 97.29%, Vision Transformer achieved 99.43%, Swin Transformer achieved the best accuracy of 99.71%, and ensemble model had a test accuracy of 98%.Explainable Artificial Intelligence techniques were integrated into the deep learning models using different packages such as Captum, Grad-CAM++, and Occlusion, to generate explanations to understand how deep learning models work and gain the trust of healthcare professionals. Moreover, a web application was designed to deploy this tuberculosis screening model in a web-based environment. The proposed method provides a fast, reliable, scalable, and cost-effective Tuberculosis detector.

Keywords: *Tuberculosis Detection, Deep Learning, Chest X-ray, ConvNeXt,Vision Transformer, Swin Transformer, Ensemble, Explainable AI, Captum.*

CHAPTER 1
INTRODUCTION

1.1 ORIGIN OF THE PROBLEM

Tuberculosis remains a serious global problem and causes millions of deaths every year due to insufficient healthcare infrastructure in developing countries. It is important to conduct timely and accurate diagnostics because it will help to prevent further spreading of the disease. Nevertheless, the use of chest X-rays for tuberculosis diagnosis has many disadvantages and requires experienced radiologists.The absence of skilled doctors in rural areas is one of the main barriers to successful diagnosis. Besides, existing diagnosis procedures take much time. In order to overcome mentioned problems, the use of automated system is proposed to detect diseases on early stages.Nowadays, the development of machine learning allowed building complex models and conducting automated analysis of medical data, which helped to create effective medical applications. Nevertheless, there is still the necessity to overcome dataset balance, high error rate, and lack of explanation of results obtained. Thus, there is a need to develop a deployable and explainable deep learning model for tuberculosis detection using chest X-rays.Besides, the use of such models becomes relevant in the conditions when demand for fast healthcare increases. The integration of intelligent systems in healthcare practice will decrease processing time and support work of healthcare professionals.

1.2 BASIC DEFINITIONS AND BACKGROUND

- **Tuberculosis:** Tuberculosis is a highly contagious infection, which occurs due to the bacterium known as Mycobacterium tuberculosis. The bacteria affect the lungs but may also infect other organs in the body. The infection is spread from one individual to another through the air droplets released when the infected individual speaks, coughs, or sneezes.
- **Chest X-ray:** Chest x-rays are imaging techniques employed in the field of medicine in which the use of electromagnetic energy results in the visualization of the chest, which includes the lungs and the heart. The technique is usually done to identify any irregularities that may exist, which include infections and various other conditions affecting the lungs and the heart.
- **Deep Learning:** Deep Learning falls under Machine Learning techniques that employ neural networks comprising of multiple layers for automatic learning of complicated patterns and representation from the vast amount of data. Deep Learning techniques are extensively used in applications like image recognition, speech recognition, and

medical diagnosis applications where they can detect features and predict outcomes without feature engineering.

- **Convolutional Neural Network:** A Convolutional Neural Network (CNN) is a deep learning algorithm designed explicitly for image analysis and feature extraction through convolutional layers. The ConvNeXt is a recent CNN model that has been developed to enhance the performance of standard CNN models through various improvements in its architecture.
- **Vision Transformer:** Vision Transformer is a deep learning architecture that classifies images by breaking them into patches and applying transformers in the form of attention techniques to identify global dependencies in the different image regions.
- **Swin Transformer:** The Swin Transformer is an architecture built on top of the concept of hierarchy with shifted window attention techniques that help in efficiently classifying images.
- **Ensemble Learning:** Ensemble learning refers to a method of combining the predictions of various classifiers or models to improve efficiency.
- **Data Augmentation:** Data augmentation involves the process of expanding an existing data set through certain techniques such as rotations, flips, scaling, and cropping, which help enhance the model's performance.
- **Class Imbalance:** Class imbalance occurs when there is a higher number of instances within one class than in another class in a given data set. This leads to bias within the data set as it becomes favoring the majority class.
- **Explainable AI:** Explainable Artificial Intelligence entails methods for making AI models explainable, revealing the relationship between the input variables and the prediction generated. In the case of radiological imaging, XAI pinpoints crucial features of the image, improving trust in model outcomes.
- **Gradcam++:**Grad-CAM++ is a cutting-edge technology used in Explainable AI to visualize those areas of the image that have an impact on a model's predictions. Grad-CAM++ achieves this by calculating gradients with respect to the target class in the feature maps produced by the last convolutional layers. The resulting heatmap is then able to highlight the areas of interest.
- **Captum:** Captum is a model interpretation library which is an open source tool developed specifically for PyTorch. This library offers many attribution methods that can be used to understand how a deep learning model generates its output. Some of these methods include Integrated Gradients, Saliency Maps, DeepLift, and Occlusion.

Using these methods, one can evaluate the influence of each feature present at the input on the final result.

- **Occlusion:** The Occlusion Method is one of the techniques in Explainable Artificial Intelligence that helps in finding important regions in an image by masking different regions in an image and analyzing how each masking will affect the output of the algorithm. The important regions are those where there will be a significant change when masked. It can be applied in chest X-rays to find important regions.

Background

Tuberculosis continues to be a major concern in the modern world, especially among developing nations whose level of access to quality medical services and facilities is not adequate. It is crucial to detect and isolate patients early enough before the disease spreads to others. However, common diagnostic techniques such as laboratory tests tend to be too costly, time-consuming, and require highly skilled experts. X-rays have been widely used to detect cases of Tuberculosis but interpreting these images has proven to be rather difficult due to subjective nature.With the recent developments in deep learning techniques, accurate diagnoses of diseases using chest X-rays is now possible and relatively affordable. Some of the recent architectures such as the ConvNeXt, Vision Transformer , and Swin Transformer possess the ability to learn advanced patterns from the input chest X-ray image data to achieve a very high degree of classification accuracy. Explainable AI will also be used in our model to increase trust in the predictions through visualization.

1.3 PROBLEM STATEMENT

The reason for the prevalence of tuberculosis disease is that there are no efficient, precise, and rapid diagnostic tools that can detect the disease quickly. For example, the interpretation of chest X-ray is still dependent on expert radiologists, which is a tedious process and requires much time. Moreover, the rising Tuberculosis cases also impose a burden on health care facilities and lead to delayed diagnosis and treatments.

Thus, an automatic and precise model that can detect the TB disease from chest X-ray images needs to be developed. In other words, it needs to overcome issues like class imbalance and have an acceptable performance level to make the results interpretable.

1.3.1 Objectives

- **Development of Deep Learning Models**

  In order to create and build high-quality deep learning systems like Swin Transformer, ConvNeXt, and Vision Transformer to detect the presence of tuberculosis from chest X-ray images through the capture of vital image features.

- **Performance Evaluation and Comparison**

  For assessing and comparing the performance of the models by analyzing their results on actual medical data using measures such as accuracy, precision, recall, F1-score, and ROC-AUC.

- **Handling Data Challenges**

  For overcoming problems like imbalanced classes and overfitting by using methods like data augmentation, class balancing during training, and early stopping to enhance the generalizability of the models.

- **Integration of Explainable AI**

  For developing explainability through the use of Explainable AI using the Captum library and methods like Grad-CAM++, Integrated Gradients, DeepLIFT, and Occlusion to produce visual explanations for lung areas contributing to the predictions made.

- **Development of Practical System**

  For designing an efficient system that can give rapid and accurate predictions with visual explanations for use in practical applications by medical professionals.

1.3.2 Outcomes

- **Accurate Tuberculosis Detection System**

  A deep learning system based on various high-performance models including ConvNeXt, Vision Transformer, and Swin Transformer was created to classify chest x-rays into the normal and tuberculosis classes accurately.

- **Comparative Model Analysis**

  Various models were compared, showing that Swin Transformer is the most accurate (accuracy = 99.71%), followed by Vision Transformer and ConvNeXt models, proving the efficiency of transformer models.

- **Improved Model Performance**

  Methods for overcoming the problem of class imbalance were employed in our work, such as data augmentation, class weights for loss function, and early stopping method.

- **Explainable AI Integration**

  Explainable AI methods via Gradcam++,Captum, such as Grad-CAM, Integrated Gradients, and Occlusion, were applied to obtain explanations from the neural network models.

- **Real-Time and Practical Deployment**

  A user-friendly application/web interface was created to predict and visualize explanations for the input images.

1.4 APPLICATIONS

1. **Medical Diagnosis Support**

   This system provides doctors with aid in diagnosing patients with tuberculosis using their chest x-rays. This is achieved by offering a second opinion to the doctor in an attempt to decrease any possible errors during the diagnosis process.

2. **Screening in Rural and Remote Areas**

   This technology can be efficiently used in rural and less developed areas where the availability of skilled radiologists is low. The use of automated analysis in such cases helps in early detection and treatment of the patient, thus reducing the spread of the disease.

3. **Hospital and Clinical Use**

   The use of this system in medical institutions, like hospitals and clinics, will help in handling patient information more efficiently. It has the potential to cut down the load on radiologists since it will be used to perform preliminary checks automatically.

4. **Telemedicine and Remote Healthcare**

   With incorporation of the system into web platforms, remote diagnostics are achieved. The patients and medical professionals will be able to upload their X-rays and receive results along with visual interpretations, which makes it very useful for telemedicine purposes.

5. **Medical Research and Training**

   The proposed structure can help scientists and students of medicine learn about disease distribution and how AI algorithms are used for diagnosis. The use of Explainable AI makes it possible to visualize the decision-making process of a machine learning algorithm, which makes it a useful educational instrument.

6. **Public Health Programs**

   The technology may be helpful for government organizations and healthcare agencies that run massive TB screening campaigns. This technology provides an efficient way of analyzing massive amounts of data, which leads to the early detection of people infected with the disease.

1.5 SUMMARY

In this chapter, we have discussed the motivation behind and background of the proposed approach to detecting tuberculosis through chest X-ray images. We have pointed out the drawbacks of the conventional method and how those have made it essential to adopt a novel and more effective solution to the problem. Specifically, our discussion included the reliance of the current methods on trained radiologists, the time required by these techniques to perform the process, and the potential for human error during the process.

Early detection of tuberculosis is crucial to controlling its spread and preventing fatal consequences among patients. In light of that, the importance of adopting artificial intelligence and deep learning to enhance the capabilities of tuberculosis detection is evident..

CHAPTER 2

REVIEW OF LITERATURE

This chapter is related to the literature review that we have collected from different articles and reviewed all the points stated in those articles.

2.1 DESCRIPTION OF EXISTING SYSTEMS

1. Improving Tuberculosis Diagnosis using Explainable Artificial Intelligence in Medical Imaging

Year: 2024

Authors: Cem Ozkurt

Observations:

In this paper, an AI approach based on deep learning for TB detection using chest X-ray imaging data, together with Explainable Artificial Intelligence (XAI) methods, was proposed. In this regard, a CNN model was utilized for image classification purposes, whereas SHAP and LIME methods were employed to make the model prediction more interpretable and meaningful. In particular, it can be observed that SHAP explains more than LIME. In conclusion, incorporating XAI into machine learning models makes the system more transparent and trustworthy[1].

2. Multi-Source Chest X-ray Datasets for Accurate and Explainable TB Diagnosis

Year: 2026

Authors: Reeba Waris Ali et al.

Observations:

The current research article presents an innovative method of fusing datasets from various sources to develop a model for detecting tuberculosis using chest x-rays. The combination of datasets from multiple sources along with the use of preprocessing strategies such as normalization, segmentation, and class balancing increases the ability of the model to generalize and minimizes bias. For developing the model, a convolutional neural network-based deep learning framework is used for classification tasks, whereas, for visualization purposes, Grad-CAM is utilized[2].

3. AI-Assisted Tuberculosis Detection using Normalization-Free Networks

Year: 2022

Authors: Vasundhara Acharya et al.

Observations:

This paper proposes normalization-free deep learning networks (NFNets) for TB detection in chest X-ray images. The network is trained with a combination of progressive resizing and enhanced augmentation strategies for better classification results. The score-CAM technique is used to explain the output through visualizations. The method provides an accurate result with a high AUC, proving that normalization-free models not only enhance performance but are stable as well[3].

4. Deployable Explainable Deep Learning System for TB Detection

Year:2026

Authors: John Agumba et al.

Observations:

In this work, a portable deep learning algorithm based on the architecture of DenseNet121 was proposed for detecting TB from chest X-ray images. This algorithm was able to incorporate Grad-CAM to offer explanations about the prediction results. The developed algorithm was deployed in both desktop and mobile environments through TensorFlow Lite. Its high performance and consistency in various testing scenarios make the developed algorithm appropriate for offline deployment in limited-resource areas[4].

5. Chest X-ray Analysis for Pneumonia and TB using CNNs and ViT with XAI

Year:2025

Authors: M. S. M. Rafi et al.

Observations:

This research article employs a two-model approach involving CNNs and Vision Transformer (ViT) models for the purpose of classifying chest X-ray scans under three different classes, namely normal, pneumonia, and tuberculosis. The use of multi-source data sets, along with some other advanced processing methodologies, is employed to ensure better generalization. In comparison to CNNs, ViT model proves to be more accurate, and saliency map approaches can explain XAI findings clinically[5].

6. Explainable Deep Neural Network for TB Detection

Year:2024

Authors: B. Uma Maheswari et al.

Observations:

In this paper, we have proposed a shallow CNN architecture for TB diagnosis from chest X-rays with an emphasis on increasing the explainability of the model. This model provides better performance results than the deep CNNs in terms of accuracy and AUC with a lower level of

complexity. Explainability methods such as CAM and LIME have been used to understand how the model works[6].

7. Systematic Review on Multimodal Learning and Explainable AI in TB Detection

Year:2025

Authors: Barbara Nansamba et al.

Observations:

The current review examines the concept of multimodal learning and the explainable AI approach to detect tuberculosis (TB). Specifically, it shows that the application of various types of data, including imaging, clinical data, and patient history, increases accuracy as compared to the use of one type of data only. The current work further indicates the relevance of using XAI due to the black box character of deep learning algorithms[7].

8. Enhanced Tuberculosis Detection Using Vision Transformers and Explainable AI with a Grad-CAM Approach

Year:2025

Authors: K. Vanitha, T. R. Mahesh, V. Vinoth Kumar, Suresh Guluwadi

Observations:

A ViT framework augmented with the use of Grad-CAM is recommended in this research paper for explainable detection of tuberculosis using chest X-rays. The proposed architecture is efficient in understanding the dependencies in an image without much pre-processing. This helped in achieving better accuracy, with the model scoring 0.98 testing accuracy and 0.97 recall score, higher than that by other models based on CNN. The visualizations provided by Grad-CAM help in understanding how important areas of lungs impact the predictions[8].

9. Tuberculosis Detection in Chest Radiograph Using CNN and Explainable AI

Year:2024

Authors: Saad I. Nafisah and Ghulam Muhammad

Observations:

This research suggests the use of a deep learning system based on CNN for automatic detection of TB from chest X-ray images. This paper explains how the use of preprocessing steps, such as lung segmentation, enhances the results because the machine pays attention only to significant portions. The use of Explainable AI helps visualize the diseased lungs[9].

10. An Explainable Hybrid Deep Learning System for Tuberculosis Detection with Grad-CAM

Year:2025

Authors: Aleka Melese Ayalew et al.

Observations:

This research presents a proposed model for TB detection using CNN, HOG, and Q-SVM. The results obtained from the proposed model show that high accuracy is attained using preprocessing, data augmentation, and feature extraction methods. The interpretation of the model has been enhanced through the use of Grad-CAM[10].

2.2 SUMMARY OF LITERATURE SURVEY

Table 2.1 Literature Review

Sno	Title and Publication details	Algorithms or technologies used	Dataset	Purpose	Remarks
1	**Year:**2025 **Title:** Improving Tuberculosis Diagnosis using Explainable AI in Medical Imaging	CNN, SHAP, LIME	Chest X-ray dataset	Improve interpretability in TB diagnosis	SHAP provides better explanations than LIME; improves trust in AI systems
2	**Year:**2026 **Title:** Multi-Source Chest X-ray Datasets for Explainable TB Diagnosis	CNN, Grad-CAM	Multi-source CXR datasets	Improve generalization using combined datasets	Dataset fusion improves robustness; Grad-CAM highlights lung abnormalities
3	**Year:**2022 **Title:** AI-Assisted TB Detection using Normalization-Free Networks	NFNet, Score-CAM	Chest X-ray dataset	Improve performance without batch normalization	Achieves high accuracy and AUC; Score-CAM provides visual explanations
4	**Year:**2026 **Title:** Deployable Explainable Deep Learning System for TB Detection	DenseNet121, Grad-CAM, TensorFlow Lite	Chest X-ray dataset	Develop deployable TB detection system	High accuracy; supports mobile/edge deployment;

					suitable for low-resource settings
5	**Year:**2025 **Title:** Chest X-ray Analysis using CNN and Vision Transformer with XAI	CNN, ViT, Saliency Maps	Multi-class CXR dataset	Multi-class classification (TB, pneumonia, normal)	ViT outperforms CNN slightly; XAI improves interpretability
6	**Year:**2024 **Title:** Explainable Deep Neural Network for TB Detection	Shallow CNN, CAM, LIME	IEEE Dataport TB dataset	Balance accuracy and interpretability	Achieved 95% accuracy; better interpretability than DenseNet; highlights lung regions correctly
7	**Year:**2025 **Title:** Review on Multimodal Learning and XAI for TB Detection	Multimodal DL, XAI techniques	Multiple datasets	Analyze existing TB detection methods	Multimodal approaches improve accuracy; XAI is essential for clinical trust
8	**Year:**2025 **Title:** Enhanced Tuberculosis Detection Using Vision Transformers and Explainable AI with Grad-CAM	Vision Transformer (ViT), Grad-CAM	Custom CXR dataset	Improve predictive performance using ViT + XAI	Achieved 0.98 accuracy, 0.97 recall; ViT captures long-range dependencies; Grad-CAM highlights important lung regions
9	**Year:**2024 **Title:** Tuberculosis Detection in Chest Radiograph Using	CNN, Explainable AI (Grad-CAM)	Chest X-ray dataset	Automatic TB detection and improved	Lung segmentation improves performance;

	CNN and Explainable AI			diagnosis accuracy	XAI provides visual explanations, enhancing clinical understanding and trust in model predictions
10	**Year:**2025 **Title:** An Explainable Hybrid Deep Learning System for Tuberculosis Detection with Grad-CAM	CNN, HOG, QSVM, Grad-CAM	Chest X-ray dataset	Improve TB detection accuracy and interpretability	Hybrid model achieves high accuracy; Grad-CAM provides visual explanations improving trust in predictions

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

2.3.1 Purpose

The goal of this software is to design an innovative and highly accurate AI-powered TB disease diagnosis from chest X-rays by leveraging various deep learning models. In this regard, ConvNeXt-Tiny, Vision Transformer, Swin Transformer, and an ensemble method are deployed to make accurate diagnoses.

Unlike conventional approaches, the main goal of this project is the use of various model interpretability techniques like Explainable AI methods such as Grad-CAM++, Captum, and Occlusion. This software will help doctors make better decisions since it will produce a heatmap highlighting the affected lungs of the patient.

The main purpose of this software is to serve as an efficient and highly accurate screening system, especially in areas where there are very few experienced radiologists.

2.3.2 Scope

It includes all stages of the process, starting from image input up until explanation generation. It is made for handling real clinical situations efficiently.

Core Capabilities

This project covers the complete pipeline:

- Input: Chest X-ray image
- Processing: Deep learning model prediction
- Output: TB / Normal classification + explanation

Key Functional Scope

- Works with binary classification (Normal vs Tuberculosis)
- Supports multiple models + ensemble comparison
- Handles imbalanced dataset (3500 Normal, 700 TB) using:
  - Data augmentation
  - Weighted loss
- Provides model evaluation and visualization

Explainability Scope

- Highlights infected lung areas using:
  - Grad-CAM++ (ConvNeXt, Swin)
  - Captum (ViT)
  - Occlusion
- Helps verify whether model focuses on clinically relevant regions

Application Scope

- TB screening in hospitals
- AI-assisted diagnosis systems
- Research in medical image analysis
- Future web-based deployment

Usability Scope

The system also includes:

- A clean and intuitive UI for clinicians
- Support for GPU acceleration for faster inference
- Logging and storing prediction results in a secure manner

The system does not replace clinical diagnosis but serves as a decision-support tool.

2.3.3 Intended Users

The software is designed for a broad range of professional and research users:

Primary Users

- **Radiologists & Pulmonologists:** To assist in quick and accurate interpretation of chest X-rays.
- **General Physicians:** For preliminary screening, especially in rural healthcare setups.

Secondary Users

- **Medical and AI Researchers:** For experimentation, dataset analysis, and performance comparisons.
- **Public Health Agencies:** To support TB eradication programs and large-scale automated screening.
- **Healthcare Software Developers:** For integration into larger hospital management or telemedicine systems.

2.3.4 Functional Requirements

FR1: Image Upload

- The system must allow users to upload single or batch chest X-ray images from local storage.
- Supported formats: PNG, JPG, JPEG, DICOM (optional extension).

FR2: Image Preprocessing

- Automatically preprocess images:
  - Resize to model-specific dimensions (224×224 or 384×384).
  - Normalize pixel values.
  - Remove noise and apply filters if needed.
- Ensure consistent quality for prediction.

FR3: Model-Based Classification

- Provide classification using:
  - ConvNeXt-Tiny
  - Vision Transformer (ViT-B/16)
  - Swin Transformer
  - Custom Ensemble Model combining the above architectures
- Support switching models dynamically through a UI dropdown.

FR4: Display Prediction Output

- Show model predictions clearly:
  - Normal
  - Tuberculosis
- Include confidence scores or probability values.

FR5: Display Performance Metrics

- Present evaluation metrics such as:
  - Accuracy
  - Precision

    - Recall
    - F1-Score
    - ROC-AUC
- Allow comparison across different models.

FR6: XAI-Based Explainability

- ➢ Generate:
- Grad-CAM++ (ConvNeXt, Swin)
- Captum (ViT)
- Occlusion maps
- ➢ Display heatmaps on X-ray

FR7: Logging & Storage

- Store prediction results including:
    - Input image
    - Prediction label
    - Confidence score
    - Date and time
- Maintain logs for clinical traceability.

FR8: Model Selection

- Users must be able to choose the desired deep learning model.
- Comparison mode should allow evaluating performance across multiple models.

2.3.5 Non-Functional Requirements

NFR1: Performance

- Inference time per image must be between 2–5 seconds on GPU.
- The system should handle a batch of images without significant delay.

NFR2: Scalability & Hardware Support

- Must support high-resolution X-rays.
- Should utilize GPU (CUDA/ROCm) for faster processing.
- Efficient memory management during large-batch inference.

NFR3: Usability

- Simple and intuitive interface suitable for clinicians.
- Clear labeling, result visualization, and minimal technical complexity.

NFR4: Security & Privacy

- Ensure patient confidentiality.
- Comply with data protection principles similar to HIPAA/GDPR.

- No unauthorized access to patient information.

NFR5: Portability

- Must run smoothly on:
    - Windows
    - Linux
- Support cloud deployment (Kaggle, Google Colab, private servers).

NFR6: Accuracy & Reliability

- Maintain >95% accuracy in validation and clinical testing scenarios.
- Outputs should be consistent and reliable.

2.3.6 Software Requirements

Implementation of the TB diagnostic tool will require a set of software packages, libraries, and platforms that would provide functionality related to deep learning, image processing, data management, and visualization.

Operating System: Windows

The project is implemented and run on the Microsoft Windows operating system, which is reliable and convenient for launching programs coded in Python. It also supports all the necessary libraries and tools used throughout the process.

Programming Language: Python

Python programming language is employed in the project as the main programming language because of its simplicity, libraries, and support communities. Python programming can effectively implement deep learning models, process, and visualize data.

Deep Learning Framework

- PyTorch – model training and inference

  The PyTorch library can be utilized to develop, train, and evaluate deep learning models, including ConvNeXt, ViT, and Swin Transformer models. This library offers dynamic computational graphs, flexibility, and GPU compatibility for efficient model training and inference.
- Torchvision – pretrained model loading

  Torchvision is employed to load pre-trained models and apply image transformations like resizing, normalization, and augmentation.

Image Processing

- OpenCV – image preprocessing

OpenCV is used for pre-processing of chest x-rays. OpenCV assists in image resizing, normalizing, filtering, and image enhancement to increase model efficiency by emphasizing important features.

Explainable AI Tools

- Captum can be utilized to produce maps that show the attribution of features in input based on approaches such as integrated gradients.
- Grad-CAM++: This approach can generate heat maps, which show important regions in the input image, helping detect infection in lungs.
- Occlusion Sensitivity: This approach allows testing the sensitivity of the input images to determine their importance for the final prediction.

Data Visualization

- Matplotlib / Seaborn – metric visualization

  Matplotlib and Seaborn: Libraries for plotting different types of graphs including accuracy curves, loss curves, confusion matrices, and ROC curves. These libraries aid in the effective analysis of model performance.

Data Handling and Processing

- NumPy, Pandas – data handling

  NumPy: This library is useful for performing various numeric operations, array manipulations, and mathematical calculations.

  Pandas: Pandas are used for organizing and processing data, including structured datasets in the form of DataFrames.

Platform for Dataset and Experimentation

- Kaggle – Dataset hosting & experimentation

  Kaggle acts as a cloud computing platform that hosts, processes, and trains machine learning models using data. The software provides a one-stop shop where the user is able to upload, manage, and use various types of datasets. Furthermore, Kaggle provides access to GPUs and TPUs for free, which greatly improves the speed at which deep learning models are trained.

2.4 SUMMARY

The literature review reveals that deep learning has evolved to become an effective technique for the diagnosis of tuberculosis through chest X-rays, with advanced models like ConvNeXt, Vision Transformer, and hybrid models demonstrating high classification performance. Several works stress the importance of applying preprocessing strategies like lung segmentation, data augmentation, and denoising to improve the model performance.

Moreover, incorporating Explainable AI methodologies like Grad-CAM, Captum, and occlusion has resolved the black box problem of deep learning by giving visual explanations about how the model works. Ensemble techniques and the use of multiple modalities have also contributed positively. From all the literature sources reviewed, it is evident that the fusion of advanced models with Explainable AI will yield accurate, transparent, and clinically relevant tuberculosis prediction systems.

CHAPTER 3
PROPOSED METHOD

The design process for the proposed Tuberculosis detection algorithm involves a systematic approach that entails several steps including data pre-processing, modeling, testing, and interpretability. The entire process is well-structured in order to guarantee accuracy, efficiency, and interpretability of the algorithm.

3.1 DESIGN METHODOLOGY

1. Data Collection

For the purposes of the experiment, we have used the following dataset: TB Chest X-Ray dataset from Kaggle. This set includes pictures taken during the diagnostic process of lung diseases.The dataset in this case includes two classes of images, namely "Normal," which refers to healthy lungs, and "Tuberculosis," which denotes infected lungs. There are a number of features of the images relevant for deep learning applications.In the current dataset, images are organized under three different categories based on their location under the main folder TB_Data. In the mentioned folder, there are three different folders named Training, Validation, and Testing. Under these folders, there are two subfolders including images from both classes. Nevertheless, the dataset is unbalanced with around 3500 images belonging to the class Normal and only 700 images in class Tuberculosis. Such an issue poses several challenges when working with such datasets since it may lead to biases during model creation. In addition, images have different sizes, resolutions, and quality because they are sourced from different places.

2. Data Preprocessing

Another critical stage during preparation for deep learning models is data preprocessing, which involves making the required changes to chest X-rays images. Given that the dataset used here includes images with different sizes and quality, the first thing done is to resize all images to have uniform size (say, 224 × 224), which is needed by models such as ConvNeXt, Vision Transformer, and Swin Transformer. This is followed by normalization of the image pixel values based on mean and standard deviation value. In order to improve the generalization capability of the model and solve the problem of imbalanced data by adding more images from the minor class (tuberculosis), image augmentation is performed for the minority class. Some of the augmentation techniques applied are random rotation, horizontal flipping, zooming, and adjusting brightness. Data augmentation will increase the dataset diversity and allow the model

to learn better features, thus minimizing overfitting.Basic preprocessing, such as noise reduction and image enhancement, may also be considered if need be.

3. Model Development

Model development involves building and training sophisticated deep learning models that will be able to correctly classify Chest X-Ray images into two classes: Normal and Tuberculosis. To do this, three novel architectures, namely, ConvNeXt-Tiny, Vision Transformer, and Swin Transformer, will be used. They were selected due to their ability to learn not only local features but also global dependencies from an input image, which is crucial for TB detection.Transfer learning will be applied when training models. Namely, pre-trained weights of the models will be fine-tuned on the dataset of the current study. Transfer learning helps in saving time and improving performance when dealing with medical data. In addition, all models will be modified, and their final layer will contain only two outputs (Normal and TB classes). In addition, proper loss functions will be used to train models; learning rates will be scheduled and early stopping will be applied.During training, models learn how to extract various kinds of information from chest X-rays, including presence of different patterns, cavities, and opacities. For example, ConvNeXt is able to extract useful spatial features while Vision Transformers can model global dependencies. Swin Transformer architecture combines advantages of previous approaches and allows efficient processing.

4. Handling Class Imbalance

However, in our case the data we will be using is imbalanced since we have more Normal images than those containing Tuberculosis. Imbalanced datasets pose the danger of the machine learning models becoming biased towards the majority class (Normal). In our case, this bias would translate to the inability of the machine learning algorithm to detect tuberculosis cases correctly. Thus, we need to use some special approaches in order to make sure that the model is capable of learning from both classes. Data augmentation is one of the techniques used in our approach. This approach involves manipulating the minority class (Tuberculosis) by applying several techniques like rotation, flipping, and brightness variation. The main idea behind this technique is that the model should be able to recognize tuberculosis even if it looks different from what it was trained on. Another method used in our approach is using a class-weighted loss function when training the model. This way, the minority class receives more attention than the majority class when computing losses.

5. Ensemble Technique

In addition to improving the efficiency and effectiveness of the tuberculosis detection system, an ensemble method is used in which the predictions obtained by several neural networks are

combined. Specifically, models such as ConvNeXt-Tiny and Vision Transformer (ViT) are used because each network is capable of detecting various features from the chest X-ray images that other networks are not capable of detecting. ConvNeXt detects local spatial features of chest x-ray images, whereas ViT detects the global relationships across the entire image.In ensemble-based machine learning models, outputs obtained by different neural networks are combined through approaches such as probability or majority voting. As a consequence, a lower level of error in decision making is expected while achieving higher accuracy of prediction since errors made by one of the models do not affect the final output.Overall, the use of an ensemble model increases the efficiency of the detection process.

6. Model Evaluation and XAI Integration

Once the training process ends, there will be an evaluation to determine the quality of the training that has been achieved. The process entails examining the performance of the model using various parameters including accuracy, precision, recall, F1 score, and ROC-AUC. This gives a perspective of how well the model performs in diagnosing cases of TB.Apart from that, the analysis of training and validation accuracy and loss curves helps in checking model learning and potential over/under-fitting problem.As part of making the system transparent and trustworthy, it is possible to use Explainable Artificial Intelligence (XAI) tools in its evaluation process. Grad-CAM++ method along with other methods such as Captum (Integrated Gradients and DeepLIFT), and Occlusion could be used to create visualizations in the form of heat maps. Heat maps indicate the most critical parts of the chest X-rays concerning model predictions related to lungs' abnormalities.The integration of XAI will allow medical staff to analyze the decision process of a neural network, which will increase its interpretability. Ensuring that the model concentrates its attention only on clinically relevant regions of the lungs is a critical requirement for an accurate diagnosis.

3.2 SYSTEM ARCHITECTURE DESIGN

The architecture proposed for tuberculosis identification is an integrated approach that takes into consideration the process of data analysis, deep learning algorithm, explainability methods, and web-based application interface. The design allows for optimal movement of data from the input stage through the process of generating predictions in Fig 3.1.

1. Input and Dataset Layer

The process starts when the chest X-rays are fed into the system as inputs, and they are acquired as part of the tuberculosis dataset. They can either be obtained from existing datasets or uploaded by users via the web application. These datasets are composed of normal images and tuberculosis-infected images that will serve as the basis of training the model.

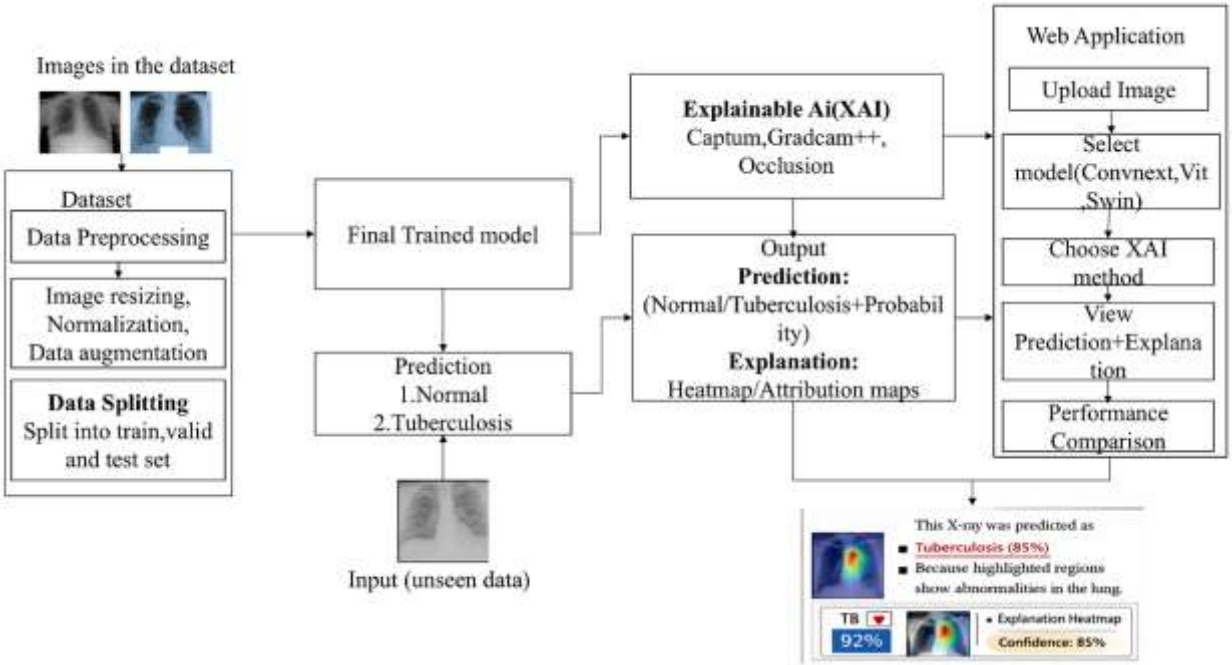

Fig 3.1 Architecture Diagram

2. Data Preprocessing Layer

After the images have been acquired, they are preprocessed in order to maintain consistency and enhance model accuracy. These include the following steps:

Image Resizing: Image size normalization to a fixed size (e.g., 224x224)

Normalization: Normalizing image pixel values for consistent training

Data Augmentation: Applying transformations like image rotation, flipping, and brightness adjustments

Data Partitioning: Partitioning the data set into train/validation/test sets

The next step after preprocessing is to create this layer of consistency before proceeding to the neural network part.

3. Model Training Layer

After performing the preprocessing process, the data is transferred to the next phase of model training:

ConvNeXt-Tiny – for capturing highly detailed spatial information

Vision Transformer (ViT) – for capturing global information

Swin Transformer – for capturing both global and local information

All the above models are trained using transfer learning techniques and fine-tuned using the TB dataset. The output of this phase is the training of a model or multiple models.

4. Prediction Layer

Upon training the model, it is applied on unseen input images to predict their class label. Given an unseen chest x-ray image:

The input image goes through preprocessing

The input image passes through the trained model

The output from the model will be:

Prediction result: Normal/Tuberculosis

Confidence score (probability)

This layer serves as the decision maker of the system.

5. Explainable AI (XAI) Layer

For increased interpretability, Explainable AI is used in the post-prediction phase:

Grad-CAM++ method - For CNN models like ConvNeXt and Swin

Captum methods - For ViT model (DeepLIFT and Integrated Gradients)

Occlusion method - Masks out the relevant region

This provides heatmaps/attribution maps for highlighting the regions responsible for the predictions made by the model.

6. Output Layer

The system has two forms of output:

1. Prediction Output:

Normal or Tuberculosis

Confidence score (for example, TB – 85%)

2. Explanation Output:

Heatmaps indicating the regions of affected lungs

Visual reason behind the prediction

These two work together to give doctors an understanding of what the model predicts and why.

7. Web Application Layer

The web-based interface is meant to ensure that the system is friendly and easy to use by everyone. The steps involved in this web application are the following:

Image Upload

Model Selection (ConvNeXt, ViT, and Swin)

XAI Technique Selection

Results Analysis

This ensures that medical practitioners can easily interact with the system without any need for technical know-how.

8. Overall Workflow

The complete system workflow can be summarized as:

Input Image → Preprocessing → Model Prediction → XAI Explanation → Output Display → Web Interface Interaction

This suggested system architecture works well to integrate deep learning and explainable AI into one system for detection of tuberculosis. This system is accurate and explainable, as well as easy to use. Modular nature of the system also ensures its further adaptation with new improvements such as portable systems.

3.3 DESCRIPTION OF ALGORITHMS

The methods and models used in the design of the TB diagnosis model are discussed in this chapter. The proposed TB diagnosis system employs sophisticated models like ConvNeXt, Vision Transformer, Swin Transformer, ensemble models, and Explainable AI.

1. ConvNeXt Algorithm

ConvNeXt is an advanced Convolutional Neural Network model based on the principles of transformer networks. The network enhances conventional CNNs through the use of large-sized kernels, depth-wise convolution, and layer normalization. ConvNeXt is able to effectively detect spatial characteristics including lung texture and diseases; thus, it can be used in the detection of tuberculosis.

Algorithm: ConvNeXt for Tuberculosis Detection

Input: Chest X-Ray(Normal/Tuberculosis)

Output: Trained ConvNeXt model for classification

Method:

1. Input and Preprocessing

- Load chest X-ray images from dataset
- Resize images to 224×224
- Normalize pixel values
- Apply augmentation (rotation, flip, brightness)

2. Patch Embedding (Stem Layer)

- Apply convolution with stride 4 to downsample image
- Convert image into feature map:

$$X = Conv(I)$$

3. ConvNeXt Blocks

Each block consists of:

a) **Depthwise Convolution**

- Apply large kernel (7×7) convolution:

$$X = DWConv(X)$$

b) **Layer Normalization**

- Normalize feature maps

c) **Pointwise Convolution (1×1 Conv)**

- Expand and reduce feature dimensions

d) **GELU Activation**

- Apply non-linearity

e) **Residual Connection**

$$Y = X + F(X)$$

4. Downsampling Layers

- Reduce spatial size and increase channel depth

5. Global Average Pooling

$$h = AvgPool(F)$$

6. Classification Head

$$P_{class} = Softmax(W_c h + b_c)$$

2. Vision Transformer Algorithm

The Vision Transformer is a deep learning algorithm that employs the concept of transformers in classification tasks on images. Unlike CNN models, which rely on convolution, the Vision Transformer splits the image into patches and treats each patch like a sequence of words in NLP tasks. The ViT is capable of modeling global dependencies between all parts of the image, making it suitable for detecting tuberculosis in chest X-rays.

Algorithm: Vision Transformer for Tuberculosis Detection

Input: Chest X-Ray(Normal/Tuberculosis)

Output: A trained Vision Transformer model for tuberculosis classification

Method:

1. Input and Preprocessing

- Load chest X-ray images from the dataset

- Preprocess each image:
  - Resize to fixed size (e.g., 224×224)
  - Normalize pixel values
  - Apply data augmentation (rotation, flipping, brightness/contrast)

2. Patch Extraction

- Divide the input image into fixed-size patches (e.g., 16×16 pixels)
- Convert image into sequence of patches:

$$I \rightarrow \{P_1, P_2, P_3, \ldots, P_n\}$$

3. Patch Embedding

- Flatten each patch and project into embedding space:

$$z_p = E \cdot Flatten(P_p)$$

4. Positional Encoding

- Add positional information to retain spatial structure:

$$z_p' = z_p + E_{pos}$$

5. Transformer Encoder Blocks

a) **Multi-Head Self-Attention (MSA)**

- Compute attention across all patches:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

b) **Feed Forward Network (FFN)**

$$FFN(x) = GELU(W_1 x + b_1)$$

c) **Residual Connections + Layer Normalization**

$$y = x + MSA(x)$$

$$z = y + FFN(y)$$

6. Classification Token ([CLS])

- Add a special classification token to the sequence
- Extract its final representation:

$$h = z_{CLS}$$

7. Classification Head

- Pass the output through a fully connected layer:

$$P_{class} = Softmax(W_c h + b_c)$$

Output

- Predicted class: **Normal / Tuberculosis**
- Confidence probability

3. Ensemble Algorithm

The ensemble approach incorporates the advantages of the ConvNeXt and the vision transformer and thus increases the accuracy and effectiveness of the tuberculous detection model. The reason is that the ConvNeXt works better for capturing local features such as the texture of lungs, whereas the Vision Transformer works for capturing dependencies throughout the whole image.

Algorithm: Ensemble Model (ConvNeXt + Vision Transformer) for Tuberculosis Detection

Input: Chest X-Ray(Normal/Tuberculosis)

Output: Final Ensemble Prediction

Method:

1. Input and Preprocessing

- Load chest X-ray images from the dataset
- Preprocess each image:
  - Resize to fixed size (224×224)
  - Normalize pixel values
  - Apply data augmentation (rotation, flipping, brightness/contrast)

2. Model Initialization

- Load pretrained models:
  - ConvNeXt-Tiny
  - Vision Transformer (ViT-B/16)
- Fine-tune both models on the TB dataset

3. Individual Model Predictions

- Pass input image through both models:

$$P_1 = Model_{ConvNeXt}(I)$$
$$P_2 = Model_{ViT}(I)$$

- Each model outputs probability scores for:
  - Normal
  - Tuberculosis

4. Probability Aggregation

- Combine predictions using average fusion:

$$P_{ensemble} = \frac{P_1 + P_2}{2}$$

5. Final Classification

- Select class with highest probability:

$$Prediction = argmax(P_{ensemble})$$

6. Output

- Predicted class:
  - Normal / Tuberculosis
- Confidence score:

$$Confidence = max(P_{ensemble})$$

4. Swin Transformer Algorithm

Swin Transformer is an architecture for Vision Transformer based on a hierarchical network structure with shifted-window multi-head self-attention operations to process images. In contrast to the vanilla transformer, which performs attention operations over the whole image, swin transformer partitions the input image into patches and performs attention operations locally to minimize computational costs. The shifted-windows mechanism ensures information transfer among different patches and enhances their representation power. Finally, a classification token aggregates global information, while the classifier performs classification by predicting the label from the learned features.

Algorithm : Swin Transformer for Tuberculosis Detection

Input: Chest X-Ray(Normal/Tuberculosis)

Output:A trained Swin Transformer model for tuberculosis Classification

Method:

1. Input and Preprocessing

- Load the chest X-ray images from the dataset.
- Preprocess each image: resize, normalize, and apply data augmentation (rotation, flip, brightness/contrast).

2. Patch Partition

- Divide the input image into non-overlapping patches (e.g., 4×4pixels each).
- Flatten each patch and project it into a feature embedding:

$$z_p = E \cdot Flatten(Patch_p)$$

3. Linear Embedding

Map each patch embedding into a feature vector of dimension C(channels):

$$z_p' = W_e \cdot z_p + b_e$$

4. Swin Transformer Blocks

a) Window-based Multi-Head Self-Attention (W-MSA)

- Perform attention inside each window:

$$Attention(Q, K, V) = Softmax(\frac{QK^T}{\sqrt{d_k}})V$$

b) Shifted Window Multi-Head Self-Attention (SW-MSA)

- Apply the same attention operation as W-MSA but with **shifted windows** to enable cross-window interaction.

c) Patch Merging (except last stage)

- Merge (concatenate) neighboring patches to reduce spatial resolution by 2 and increase channel dimension:

$$[x_1, x_2, x_3, x_4] \;\Rightarrow\; W_m \cdot [x_1, x_2, x_3, x_4]$$

5. Classification Head

- Apply global average pooling to the final feature map to obtain a feature vector:

$$\mathrm{h} = \mathrm{AvgPool(F)}$$

- Pass the pooled feature through a fully connected layer and apply softmax:

$$\mathrm{P_{class}} = \mathrm{Softmax(W_c h + b_c)}$$

5. Explainable AI Captum

Captum is an open-source library built on PyTorch that aims at interpreting the outputs of deep learning models by assigning attribution scores to their inputs. It includes visualization methods such as Grad-CAM, Integrated Gradients, and DeepLIFT to emphasize the areas most influential in the decision-making process of the model. Using Captum, doctors can comprehend how a model makes its decision regarding whether a disease is Tuberculosis or Normal using chest X-rays.

Captum for Model Explainability

Input:

Trained deep learning model (ConvNeXt / ViT / Swin Transformer)

Chest X-ray images (Normal / Tuberculosis)

Output:

Attribution maps or heatmaps showing important regions influencing predictions

Method:

1.Select Model and Test Data

- Choose a trained model for TB detection.
- Use test images that the model has not seen during training.

2. Choose Captum Attribution Method

Example options:

- Integrated Gradients
- Saliency
- DeepLift

3.Compute Attribution Maps

- Run forward pass to get prediction for the image.
- Compute attributions:
- **Integrated Gradients:**

$$\text{Attribution}(x) = (x - x') \times \int_{\alpha=0}^{1} \frac{\partial F\left(x' + \alpha(x - x')\right)}{\partial x} d\alpha$$

- where x is input, x' is baseline, F is model output, α: a scaling factor between 0 and 1
- **Saliency:**

$$\text{Attribution}(x) = \left|\frac{\partial F(x)}{\partial x}\right|$$

4.Visualize Attribution

- Overlay the attribution/heatmap on the original X-ray image.

5.Interpret and Present Results

- Highlight regions influencing the prediction.
- Display predicted class + probability with the heatmap.

6.Explainable AI Gradacam++

Grad-CAM++ is a type of Explainable AI (Explainable Artificial Intelligence). It visualizes the crucial parts of an image on which the model prediction depends. Grad-CAM++ is an improvement over Grad-CAM because it performs better at localization and can detect several crucial parts. Grad-CAM++ is particularly effective in explaining predictions of CNN models

such as ConvNeXt and Swin Transformer, as it visualizes the lung area to diagnose tuberculosis.

Grad-CAM++ for Model Explainability

Input:

- Trained deep learning model (ConvNeXt / Swin Transformer)
- Chest X-ray images (Normal / Tuberculosis)

Output:

- Heatmaps highlighting important regions influencing predictions

Method:

1. Select Model and Test Data
   - Choose a trained CNN-based model
   - Use unseen test images
2. Identify Target Layer
   - Select the last convolutional layer of the model
   - This layer contains spatial feature information
3. Compute Gradients
   - Perform forward pass to get prediction
   - Compute gradients of target class with respect to feature maps

$$\frac{\partial y^c}{\partial A_{ij}^k}$$

4. Compute Weights (Grad-CAM++)
   - Calculate importance weights using higher-order gradients:

$$\alpha_{ij}^k = \frac{\partial^2 y^c}{\partial (A_{ij}^k)^2}$$

   - Final weights:

$$w_k^c = \sum_i \sum_j \alpha_{ij}^k \cdot ReLU\left(\frac{\partial y^c}{\partial A_{ij}^k}\right)$$

5. Generate Heatmap
   - Combine weights with feature maps:

$$L_{GradCAM++}^c = ReLU\left(\sum_k w_k^c A^k\right)$$

6. Visualize Attribution

- Upsample heatmap to original image size
- Overlay on chest X-ray

7. Interpret Results

- Highlight lung regions contributing to TB detection
- Display prediction + heatmap

7.Explainable AI Occlusion

Occlusion is one of the many Explainable AI methods that do not depend on any specific model and help identify the areas of the picture that are most valuable to the model. Occlusion masks parts of the picture and analyzes the difference in model prediction before and after doing so.

Occlusion for Model Explainability

Input:

- Trained deep learning model (ConvNeXt / ViT / Swin Transformer)
- Chest X-ray images (Normal / Tuberculosis)

Output:

- Importance map showing critical regions

Method:

1. Select Model and Test Data

- Choose trained model
- Select test image

2. Define Occlusion Window

- Select patch size (e.g., 15×15 pixels)
- Choose stride for sliding window

3. Slide Occlusion Window

- Move window across image
- Mask region (set to zero or baseline):

$$I_{occluded} = I \odot M$$

4. Compute Prediction Change

- For each occluded image:

$$\Delta = F(I) - F(I_{occluded})$$

5. Generate Importance Map

- Assign importance score to each region based on change
- Larger change → higher importance

6. Visualize Attribution

- Construct heatmap from importance scores
- Overlay on original X-ray

7. Interpret Results

- Identify regions critical for TB prediction
- Display prediction + explanation

3.4 DESCRIPTION OF DATASET AND TOOLS

3.4.1 Dataset Description

The Tuberculosis Chest X-ray dataset is the data set that will be employed in this study. The dataset is freely available on the Kaggle website. This particular dataset is designed to provide data for developing computer vision models that can automatically diagnose TB based on chest X-ray imaging. The chest X-ray images contained within this data set are posteroanterior (PA) view images.

The data set comprises two main classes:

Normal – These images represent healthy individuals with no radiological signs of Tuberculosis.

Tuberculosis – These images belong to patients confirmed to have TB infection, typically showing lesions such as infiltrates, nodules, consolidations, or cavitary regions.

Overall, the number of images contained within the dataset is about 4,200, out of which there are approximately 3,500 Normal images and about 700 Tuberculosis images. Consequently, there is an imbalanced ratio between these classes, as the amount of images in the former category is almost five times bigger than in the latter one. Such data imbalances pose a serious challenge during medical imaging classification, because deep learning algorithms can be trained to favor predicting the majority classes, resulting in poor recall rates for the minority classes like TB.

Several techniques have been employed to cope with this problem:

Class-Weighted Loss Functions:

The cost function was modified such that higher weights were assigned to the minority class. This will ensure that any error made in predicting the TB class is penalized more than the error made for predicting the non-TB class.

Data Augmentation for Minority Class:

Further augmentation methods including random rotations, flips horizontally, brightness modifications, and minor zooming were done with increased vigor on the TB dataset. This is done to increase the variation artificially and avoid overfitting to the small TB dataset.

Balanced Sampling:

When training the model, samplers were employed to ensure that each batch had an equal number of Normal and TB images. This ensured that the model could discriminate between both classes.

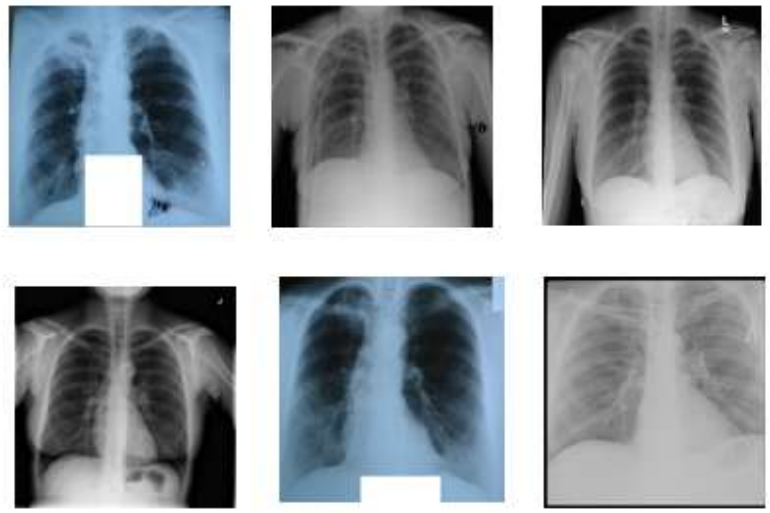

Fig 3.2 Sample Images in the dataset

Chest X-ray Images of Normal Patients and Tuberculosis-Positive Patients Fig 3.2 provides some examples of chest X-ray images available in the dataset, displaying Normal and Tuberculosis-positive images. It can be observed that there is considerable variability between images in terms of their contrast, brightness, orientation, and general quality. Normally, chest X-rays should provide a clear depiction of lung fields, defined anatomical landmarks, and the absence of any pathological changes. Such variability is due to the diversity of patients being imaged as well as the diversity of the machines used for imaging. Tuberculosis images usually contain some radiological signs associated with pulmonary tuberculosis: opacities, infiltrations, nodular formations, and consolidations. The sizes, densities, and locations of abnormal areas vary in each case.It provides another proof of the complexity of the task at hand, illustrating the need to implement robust deep learning models.

Dataset splitting for training, testing, and validation models

To ensure that the deep learning algorithms utilized in this study (ConvNeXt, Vision Transformer, Swin Transformer, and Ensemble) are adequately trained and tested, the dataset consisting of chest x-rays is divided into three parts: training, validation, and testing. Dataset splitting is an essential step in all machine learning processes since it is used to assess the model's generalization capability.

In this case, the dataset is split as follows:

- **Training Set –** 70%

- **Validation Set –** 20%
- **Test Set –** 10%

These percentages ensure a balance between providing enough data for learning while retaining sufficient examples for tuning and final evaluation.

Purpose of spilt

Training Set: The set where the model learns about the patterns and features of the data. From this data, the model reduces the loss by improving the accuracy through weight adjustment.

Validation Set: It is used for the optimization of the parameters of the model during the training process and the assessment of its performance during the training phase. This means that through this, you will be able to test the model's ability on unseen data, and hence identify any case of overfitting or underfitting and make the necessary adjustments to optimize performance.

Test Set: The test set is used only once at the end of the training of the model when there is no need to train further. It is done so as to test the performance of the model on the unseen data.

3.4.2 Tools and Technologies used

This tuberculosis detection model system will be developed through a blend of some sophisticated programming languages, frameworks, software libraries, and cloud computing technologies. These were specifically selected to ensure efficient development, optimal computing capacity, scalable processes, and medical imaging compatibility.

Programming Language:

The software development process of the entire system involved Python as the main programming language.

Important benefits of using Python in the project are as follows:

- The possibility to quickly prototype and test various deep learning algorithms
- Very good documentation and help available from a large Python community
- Easy integration with libraries for implementing deep learning algorithms and for science
- Good compatibility with GPU-based computation environments

With Python, the whole process from data preprocessing to algorithm visualization was implemented.

Deep Learning Framework:

PyTorch was chosen as the primary deep learning platform for developing and training the model. The dynamic computation graph offered by PyTorch is ideal for research projects

because it facilitates flexibility in designing the model and makes debugging more straightforward.

The advantages of using PyTorch include:

- The availability of pretrained models like ConvNeXt, Vision Transformer (ViT), and Swin Transformer
- Simple customizability of neural network layers and loss functions
- Fast GPU computation through CUDA
- Simple incorporation of explainable AI techniques, such as Captum

With PyTorch, the pre-trained models were fine-tuned with transfer learning, enabling the system to produce excellent results despite the small medical data set.

Supporting Libraries and Tools:

To support data handling, preprocessing, visualization, and evaluation, several auxiliary libraries were used.

- **Torchvision:**

  Torchvision was used for handling image-related operations and model utilities. It provides:

  Standard datasets and pretrained vision models

  Image transformation and augmentation techniques such as resizing, rotation, flipping, and normalization

  Efficient data loading using DataLoader

  In this project, Torchvision played a critical role in data augmentation, which helped reduce overfitting and improve model generalization.

- **OpenCV:**

  OpenCV was used for low-level image preprocessing tasks. It supports:

  Image resizing and format conversion

  Pixel-level image manipulation

  Noise reduction and enhancement operations

  OpenCV ensured that chest X-ray images were uniformly processed before being passed to the deep learning models.

- **NumPy & Pandas:**

  NumPy and Pandas were used for efficient data manipulation and numerical computation:

  NumPy enabled fast matrix operations and numerical calculations

  Pandas was used for handling labels, metadata, and performance logs

These libraries simplified dataset organization, result storage, and metric computation.

- **Matplotlib / Seaborn:**

  Matplotlib and Seaborn were used for visualization and performance analysis. They helped in:

  Plotting training and validation accuracy and loss curves

  Visualizing confusion matrices

  Analyzing model convergence behavior

  These visualizations were essential for model evaluation, debugging, and result interpretation.

- **Scikit-learn:**

  Scikit-learn was used for computing evaluation metrics and validating model performance. It provided:

  Accuracy, precision, recall, and F1-score calculations

  ROC curves and AUC computation

  Confusion matrix generation

  These metrics ensured a quantitative and unbiased evaluation of the tuberculosis detection system.

- **Kaggle / Google Colab:**

  Kaggle and Google Colab were used for model training and experimentation due to their free GPU support and cloud-based execution.

  Their advantages include:

  Access to NVIDIA GPUs for faster training

  No requirement for local high-end hardware

  Easy dataset management and collaboration

  Reproducible experimentation through notebooks

  Kaggle was primarily used for structured dataset handling and training, while Google Colab supported quick experimentation and testing.

3.5 SUMMARY

This proposed method is a computer-assisted system that detects tuberculosis based on images captured via chest X-rays using deep learning techniques. The first step is to preprocess the data by resizing the image dimensions to be uniform, applying normalization to the pixels, and enhancing them using augmentation such as rotations, flips, and rescaling. This ensures better generalizability of the system. Due to class imbalances within the dataset, loss weighting and balance sampling are employed to allow for effective learning of the samples from both classes.

Transfer learning is used through the use of pre-trained models such as ConvNeXt, ViT, and Swin Transformer, where the layers are fine-tuned to perform binary classification.

The efficiency of the proposed solutions is assessed on the basis of several performance metrics, such as accuracy, precision, recall, F1-score, and ROC-AUC. An additional step to further improve the quality of predictions is to apply the concept of ensembling to combine multiple models' outputs and create a more robust prediction algorithm. Moreover, in order to facilitate interpretability of AI solutions, several Explainable AI approaches, such as Grad-CAM++ and Integrated Gradients (Captum), as well as Occlusion Sensitivity, have been applied to visualize and interpret clinically significant parts of the X-ray image.

CHAPTER 4
RESULTS AND OBSERVATIONS

In this chapter, the results obtained through the deployment of the proposed tuberculosis detection model are presented and analyzed extensively for understanding model performance. In particular, the performance of the models that have been designed including ConvNeXt, Vision Transformer , Swin Transformer and the combined model will be discussed in terms of performance measures such as accuracy, precision, recall, F1-Score, and ROC-AUC.Apart from analyzing the performance quantitatively, the model has been analyzed qualitatively as well using the methods such as Explainable Artificial Intelligence including Grad-CAM++ Captum, and Occlusion. The objective of qualitative analysis is to understand which portions of the chest X-ray image play a role in the decision-making process of the proposed deep learning model.

4.1 DESCRIPTION OF RESULTS

The results for the proposed system demonstrate that the deep learning models can precisely categorize chest X-ray images based on their classes as either normal or Tuberculosis. The analysis is done using the confusion matrix and some metrics including accuracy, precision, recall, F1 score, and ROC-AUC. The results confirm that the models have high reliability. Of all the models used, the Swin Transformer has the highest performance, followed by the Vision Transformer, then ConvNeXt, and finally the ensemble model which has stability in its predictions. Besides, Explainable AI methods can produce heatmaps to emphasize vital regions in the lungs.

Confusion Matrix:

The confusion matrix is an essential tool for evaluating machine learning models. It displays a tabulated form of comparison between actual classes and predicted values from the classifier. It comprises four elements; true positive (TP), true negative (TN), false positive (FP), and false negative (FN). In tuberculosis prediction, TP refers to instances where the disease was accurately detected, TN refers to instances where the absence of the disease was accurately predicted, FP refers to instances where there is no disease but is diagnosed as having tuberculosis, and FN refers to instances where there is tuberculosis but the model predicts otherwise.

Performance Metrics

Metrics are quantitative criteria used to measure the level of success that the ML/Deep Learning algorithm has achieved while executing the specific task. In the case of a problem such as the prediction of tuberculosis infection, performance metrics are employed to determine whether the machine learning or deep learning algorithm is capable of making predictions correctly by comparing actual and predicted classes (normal and TB). Popular performance metrics for such algorithms include Accuracy, Precision, Recall, F1 score, and ROC-AUC among others.

1. Precision:

Precision is a metric that evaluates the precision of the predicted positive results of the algorithm. In other words, it shows how many of the patients classified as having TB have TB indeed. The higher the precision, the fewer the number of false positives.

$$Precision = \frac{True\ positive}{true\ positives + false\ positives} \quad \text{-(Eq 1)}$$

2. Accuracy:

Accuracy is a measure of the correctness of the model as it measures the ratio of correct predictions made by the model (TB and Normal). Although accuracy offers an insight into the overall model performance, it cannot suffice on its own, especially for imbalanced datasets.

$$Accuracy = \frac{True\ positives + True\ Negatives}{Total\ Instances} \quad \text{-(Eq 2)}$$

3. Recall (Sensitivity):

Recall indicates how good the classifier is at finding all the true cases of tuberculosis. Recall becomes extremely important when the model is applied in the field of healthcare because there may be severe repercussions if even a single case of tuberculosis goes undetected.

$$Recall = \frac{true\ positives}{True\ positives + False\ negatives} \quad \text{-(Eq 3)}$$

4. F1-Score:

F1 Score is the harmonic mean of the two metrics mentioned and is used as an evaluation metric when class imbalance is present since it takes into account false positives and false negatives.

$$F1\text{-}score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad \text{-(Eq 4)}$$

5. ROC Curve and AUC:

ROC (Receiver Operating Characteristic) Curve

This is a graph that provides information on how well the binary classifier works at various thresholds. This graph compares the False Positive Rate and True Positive Rate (Recall). AUC

(Area Under Curve) represents the graph in one figure. AUC shows how well the model performs; an AUC closer to 1 implies that the model performs very well. AUC provides information on whether the model can differentiate between normal and tuberculosis.

$$\text{AUC}=\int_0^1 TPR(FPR)\, d(FPR) \qquad \text{-(Eq 5)}$$

4.2 RESULTS

The accuracy levels for the results obtained from the proposed tuberculosis detection system indicate that all the deep learning models implemented have achieved good accuracy when it comes to classification of the chest X-rays into either Normal or Tuberculosis classes. From among the three models implemented, the one with the highest accuracy level is Swin Transformer, followed by Vision Transformer and then ConvNeXt.

ConvNeXt Model

ConvNeXt training and testing were conducted using the Tuberculosis Chest X-ray Dataset for classifying Normal and Tuberculosis classes. ConvNeXt performed well because of its advanced convolutional neural network design and good features extraction capability.

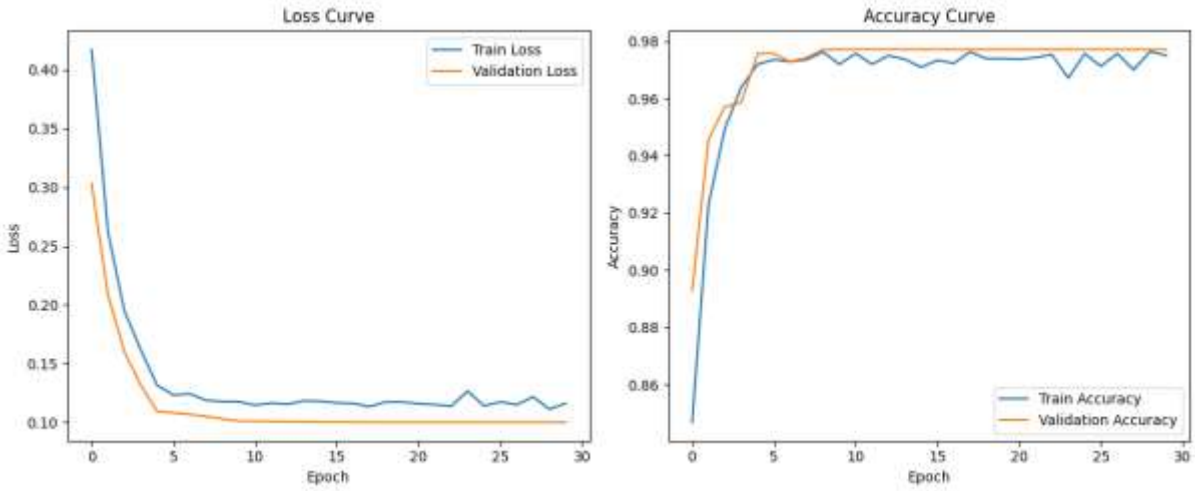

Fig 4.1 Accuracy and Loss Graphs for ConvNeXt model

As seen from Fig 4.1, ConvNeXt shows excellent results and stability due to successful training. From the loss chart, one can see a rapid decline in losses for both training and validation in the first epoch: from about 0.42 and 0.30 to almost 0.10, where the losses become stable without any significant changes. As seen from the similarity in the loss charts, overfitting is not observed here, and the model works well even on validation data. In the same way, a rapid improvement in accuracy can be observed for training from about 85% to 97–98% and the same value is achieved for validation and stays at the same level all through the process. Again, the small difference between training and validation proves the high effectiveness of

the model. Thus, the proposed model successfully classifies tuberculosis cases with high accuracy and fast convergence rate.

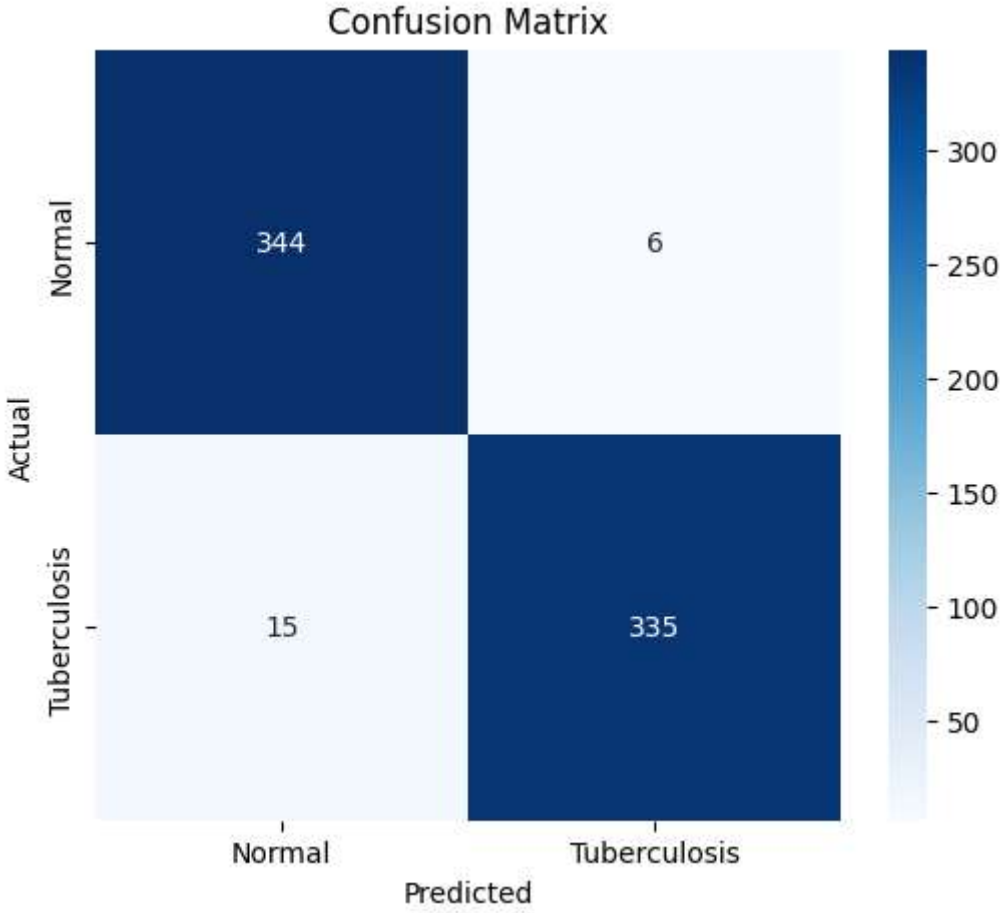

Fig 4.2 Confusion Matrix of ConvNeXt model

The Confusion Matrix of the ConvNeXt model used to detect tuberculosis is illustrated in Fig 4.2. As depicted from the matrix, there are some observations that can be made concerning the model's performance in classifying chest X-ray images as either Normal or Tuberculosis. Out of all the predictions made by the model, it correctly detects 344 normal cases as Normal, which can be referred to as True Negatives. Furthermore, out of all the images classified by the model, it successfully predicts 335 tuberculosis cases as Tuberculosis, which can be termed as True Positives. Only a few mistakes are observed in predicting tuberculosis and normal cases where six normal cases were predicted as Tuberculosis and 15 tuberculosis cases were predicted as Normal. From the results shown in the confusion matrix, the ConvNeXt model shows high accuracy and discrimination ability.

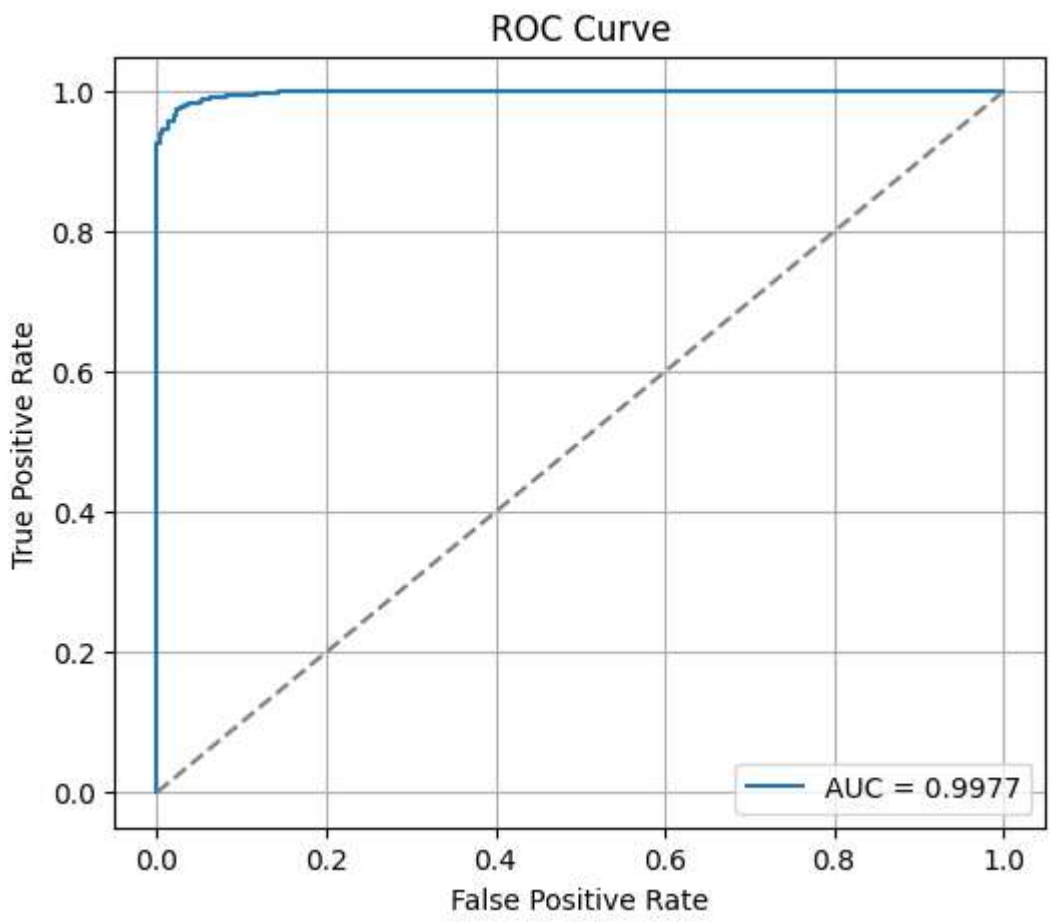

Fig 4.3 ROC of ConvNeXt model

Fig 4.3 depicts the Receiver Operating Characteristic (ROC) plot of the ConvNeXt model for classifying tuberculosis disease. In the ROC plot, there is an inherent relationship between the true positive rate (sensitivity) and false positive rate based on different cut-off values. The plot is seen to lie closer to the top left corner, thereby reflecting very good classification efficiency. The area under the curve (AUC) score obtained by the ConvNeXt model is seen to be quite high, around 0.9977, which indicates excellent separation between the Normal and Tuberculosis categories. Furthermore, the ROC plot lies far above the diagonal, thereby proving that the model is definitely better than random classification.

Vision Transformer Model

The Vision Transformer model was tested to determine how effective it is in classifying chest X-rays into the Normal and Tuberculosis classes. The findings show that the Vision Transformer model is effective in learning from the data to provide accurate predictions. Different evaluation methods and visual representations were employed to determine the performance of the model.

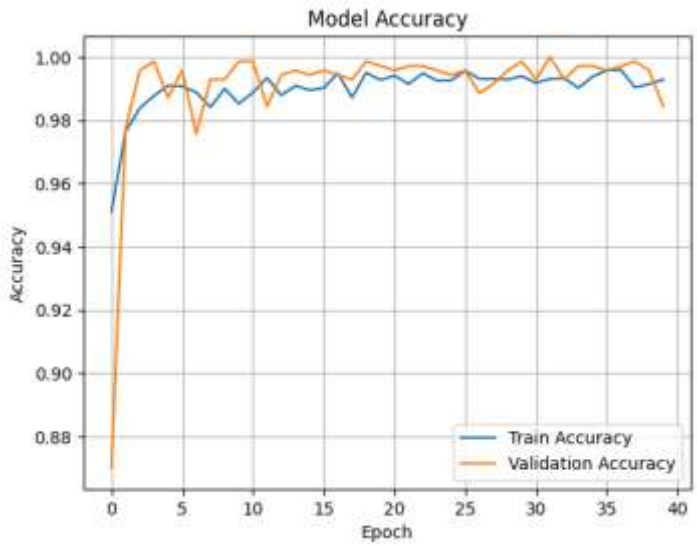

Fig 4.4 Accuracy graph for Vision Transformer

Fig 4.4 illustrates an outstanding performance based on the accuracy curve graph. From the training accuracy plot, there is a gradual rise from 95% to 99%, signifying effective feature extraction from the images of chest X-ray. Likewise, the validation accuracy also has an upward trend from the start up to about 98-100%. This indicates excellent generalization. In addition, from the similarity between training and validation curves, it can be seen that there is no problem of overfitting in the model used. However, slight variations are present in the validation accuracy, which is expected when conducting model training. The model attains almost optimal accuracy in the early training epochs.

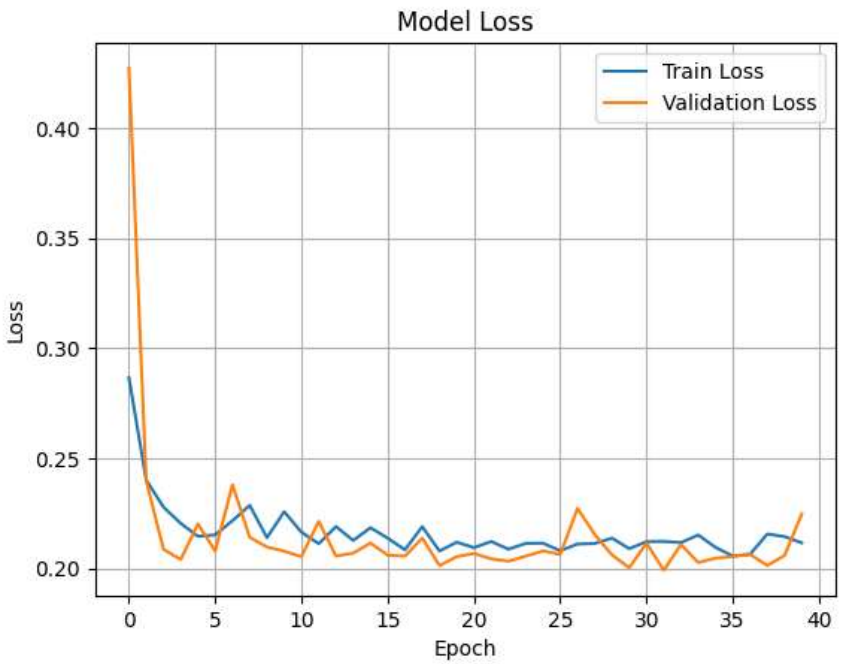

Fig 4.5 Loss graph for Vision Transformer

The loss function of the Vision Transformer model during the training and validation processes is shown in Fig 4.5. As can be seen from the figure, the loss value of the training set gradually decreases from about 0.28 to approximately 0.21, suggesting that the model learns well from the data. Likewise, the loss value of the validation set rapidly declines from around 0.43 to approximately 0.20 in the early stages, indicating quick convergence. Once past the sharp descent phase, the values of the training and validation losses remain fairly stable with some variations, indicating consistent learning by the model. Since the curves are quite close to each other, there is little evidence of overfitting or underfitting on the part of the model.

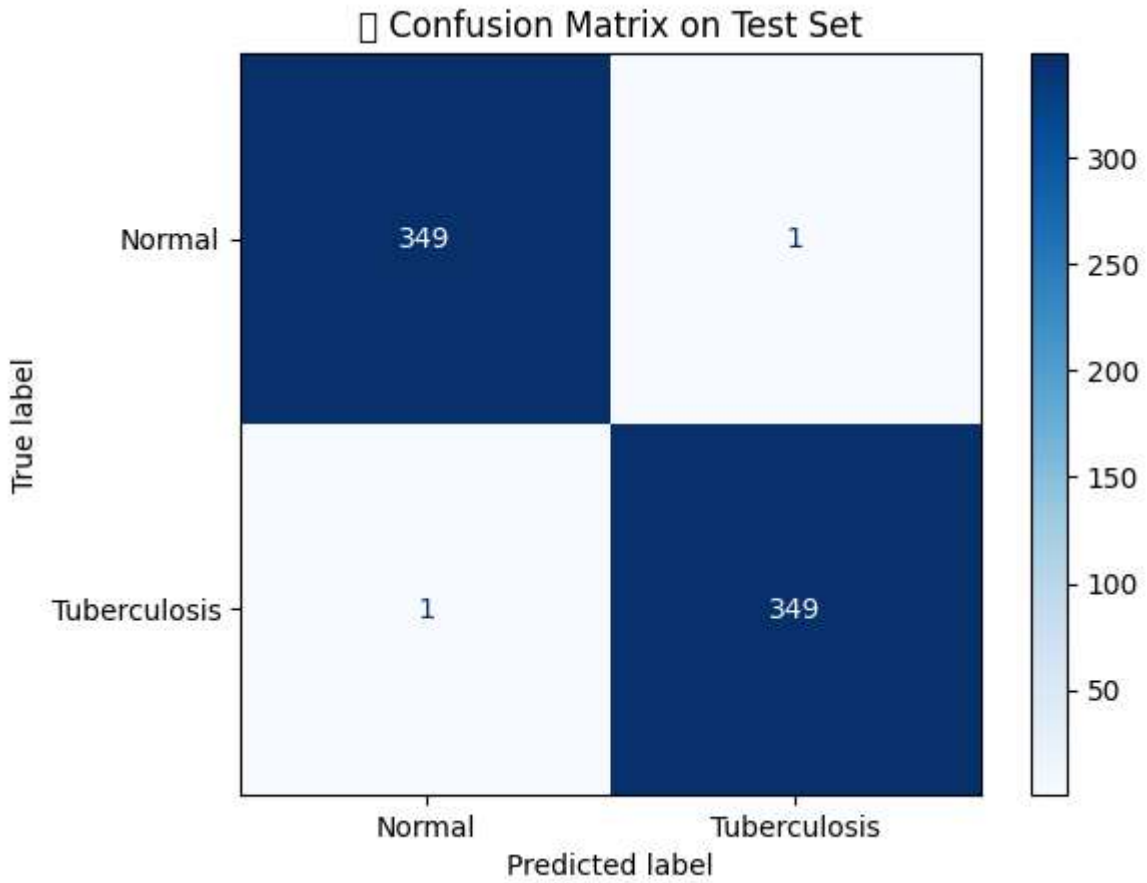

Fig 4.6 Confusion Matrix for Vision Transformer

The confusion matrix of the Vision Transformer model based on the test dataset is illustrated in Fig 4.6. From the table, it can be seen that there were 349 instances classified as True Negatives (TNs) and 349 cases that have been properly classified as True Positives (TPs). The Vision Transformer model produced negligible errors since it failed to classify only 1 normal instance as Tuberculosis (FP) and 1 tuberculosis instance as Normal (FN). Based on the outcomes obtained, it can be seen that the Vision Transformer model is highly accurate and can effectively distinguish between the two classes. This can be particularly useful when applied in medical diagnostics due to the low number of FN classifications.

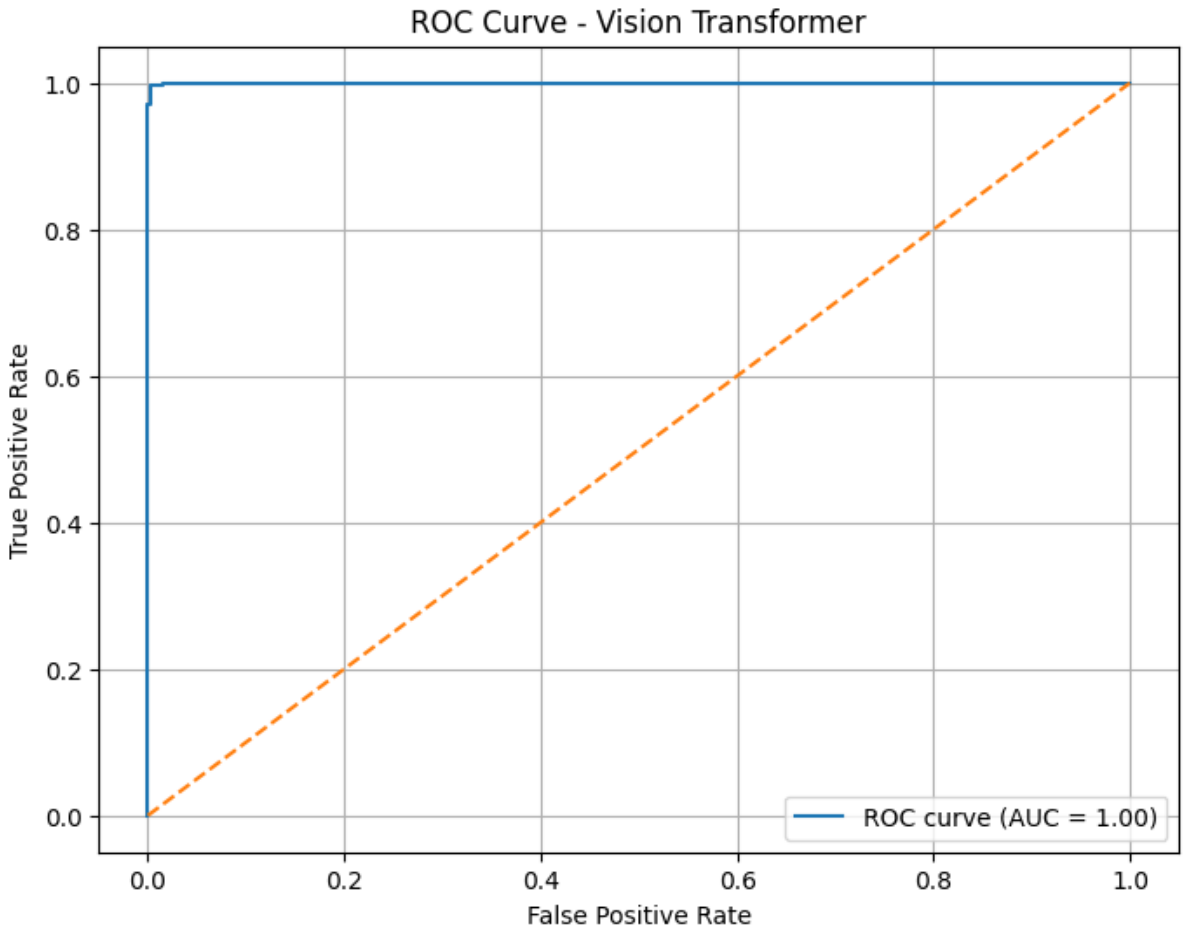

Fig 4.7 ROC Curve for Vision Transformer

Fig 4.7 depicts the ROC curve of the Vision Transformer for the detection of tuberculosis disease. The ROC curve represents the trade-off between the true positive rate and the false-positive rate at various thresholds. The ROC curve of the Vision Transformer nearly coincides with the upper left region of the graph, implying perfect classification results. The area under the curve (AUC) score obtained by the Vision Transformer is 1.00, which reveals that the classifier exhibits remarkable discrimination capability between the Normal and Tuberculosis classes. Furthermore, the ROC curve lies far away from the reference diagonal line, ensuring that the classifier is much more superior to the random classifier.

Ensemble Model

The proposed hybrid model, which uses ConvNeXt and Vision Transformer, proves to be a better model in classifying tuberculosis. It is observed that it achieves extremely high accuracy along with higher precision and recall values, which ensure that there is no false detection of tuberculosis patients or any other problem.

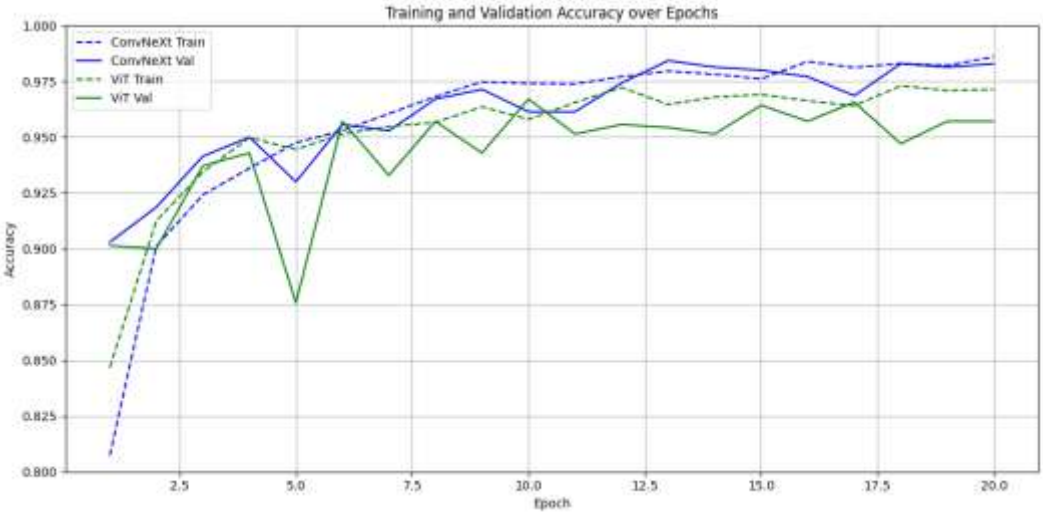

Fig 4.8 Accuracy and Loss Graphs for Ensemble model

Accuracy during training and validation of the combined model is depicted in Fig 4.8 where the convolutional model ConvNeXt is used in conjunction with the vision transformer called ViT. The figure clearly depicts how both ConvNeXt and ViT provide a continuous increase in accuracy over the different epochs, although ConvNeXt provides slightly better results compared to ViT. It can be seen how close the validation accuracy results are to the training accuracy, hence, no overfitting takes place and the generalization process is done properly. Although some small changes occur in ViT validation accuracy, the overall results are constant and accurate at every epoch.

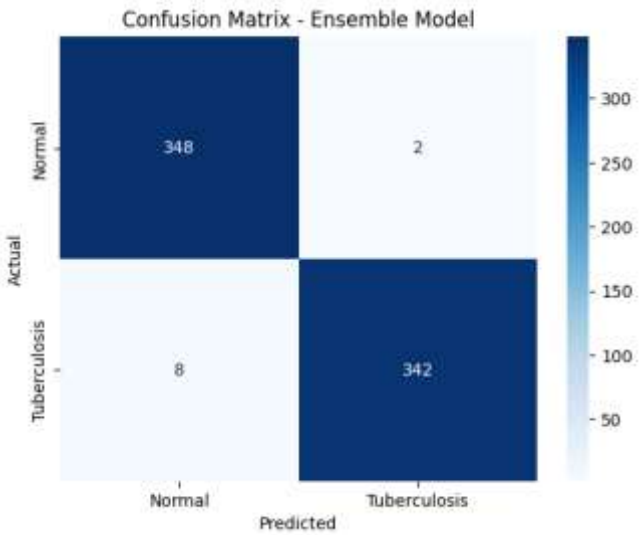

Fig 4.9 Confusion Matrix for Ensemble model

Confusion matrix of the ensemble algorithm combining ConvNeXt and Vision Transformer models for the classification of tuberculosis is shown in Fig 4.9. According to the matrix, 348 normal cases are classified as Normal (TN) and 342 cases of tuberculosis are recognized as

Tuberculosis (TP). Thus, there are few mistakes, including the two false positives where normal images were classified as TB. On the other hand, there are 8 false negatives when images of tuberculosis are mistaken for normal images. The low number of mistakes in terms of false negative suggests good classification performance and capability of the ensemble model used in the analysis.

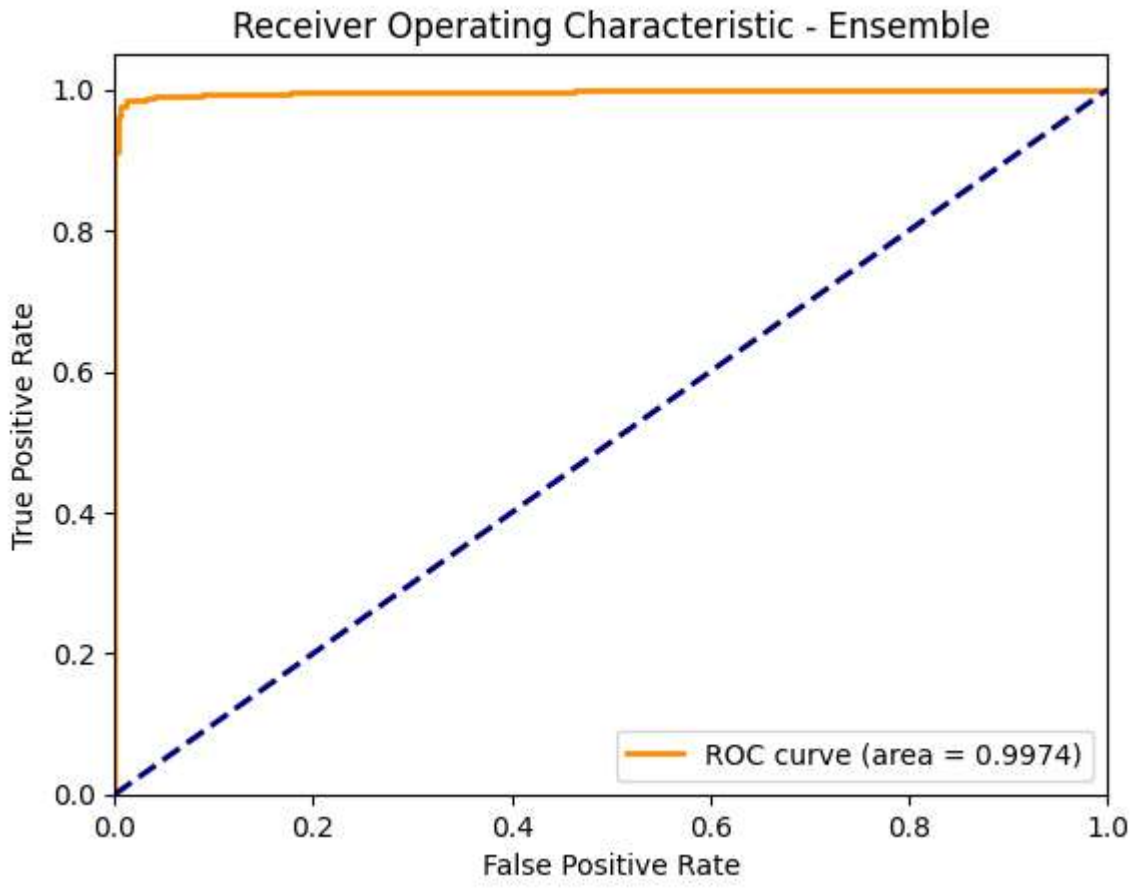

Fig 4.10 ROC Curve for Ensemble model

The ROC curve of the proposed model, which uses the combination of ConvNeXt and Vision Transformers for classifying tuberculosis, is depicted in Fig 4.10. As seen from the figure, the ROC curve is located very close to the upper left corner, suggesting a high accuracy of the developed model in performing its tasks. Moreover, the AUC value of the model is about 0.9974, which shows a high ability of the classifier to distinguish the two categories – Normal and Tuberculosis. Furthermore, the developed model outperforms the classifier used at random since its ROC curve is far away from the diagonal line.

Swin Transformer

The classification task that was carried out on the Chest X-ray image dataset using the Swin Transformer model is to classify the images into two classes namely Normal and Tuberculosis.

This analysis shows how well the model is capable of extracting local and global features using its transformer structure.

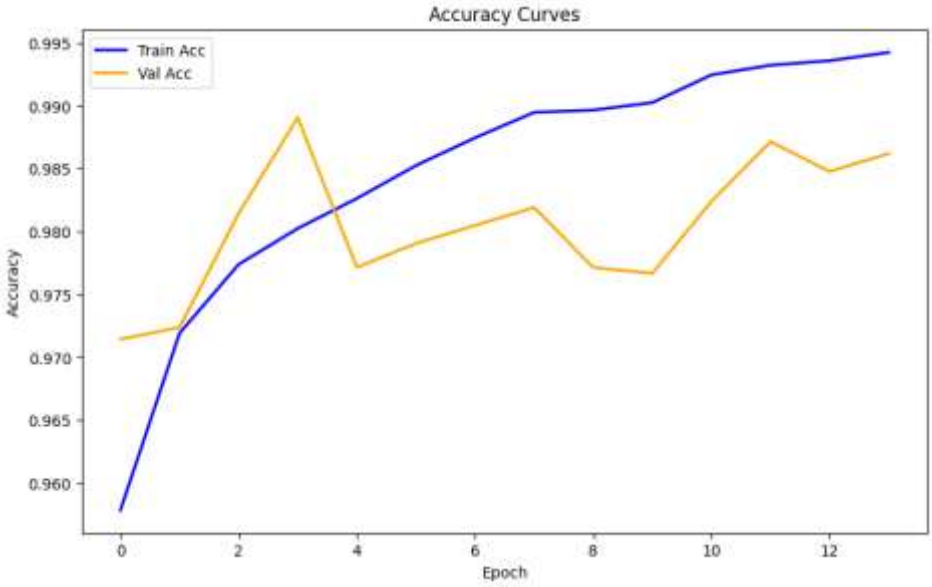

Fig 4.11 Accuracy curves of Swin Transformer

Fig 4.11 presents the accuracy curves for both the training and validation processes throughout the complete training period of the proposed model. These curves can help provide insights into the ability of the model to learn discriminative features from the chest X-ray image dataset. It is worth noting that the accuracy for training starts at around 0.958 for the first epoch and then maintains an increasing trend until approaching about 0.994 in the final epoch. It means that the training model has a good performance in learning new features when it gets more data. In the same manner, there is a clear positive trend in the validation accuracy. This starts with a steep increase at the beginning, from approximately 0.972 to almost 0.989. After the initial steep increase, the validation accuracy experiences slight ups and downs, but overall, it stays consistent within the range of 0.977 to 0.988. Minor deviations in the graph are expected because of the complex nature of the dataset used in the experiment.

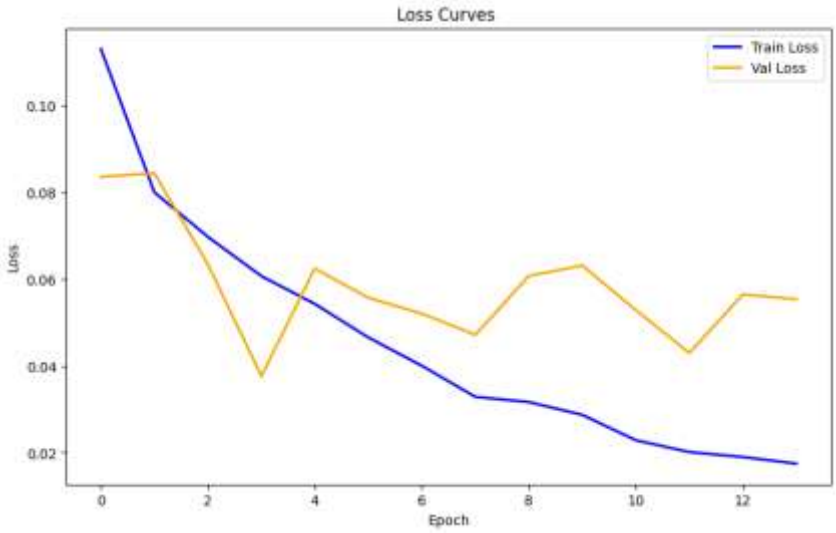

Fig 4.12 Loss curves of Swin Transformer

The Fig 4.12 presents the graph of training loss and validation loss through the training epochs, showing the behavior of the optimization process of the model. The training loss drops quickly from around 0.11 to 0.017, meaning that the model has learned well during the training process. While the validation loss shows a decreasing trend at first, which means that the generalization capability of the model improves, it experiences some fluctuations towards the end of training. However, the fluctuation does not seem to affect the validation loss significantly, and it stays relatively constant at 0.04 to 0.06.

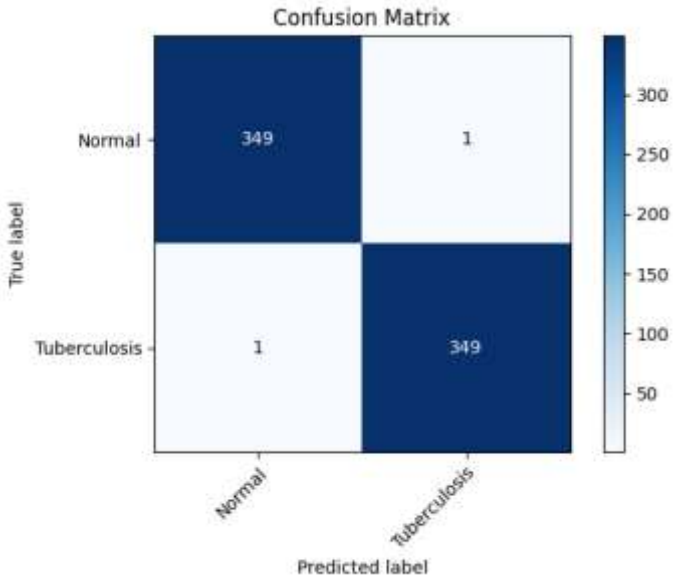

Fig 4.13 Confusion matrix of Swin Transformer

The confusion matrix for the classification result achieved by the model on the test set can be seen in Fig 4.13. The model was able to correctly classify 349 cases of normal images and 349 cases of tuberculosis images. Thus, it was evident that the model had a high capacity to

discriminate both classes. Moreover, there were only two errors made by the model: one case of normal image being classified as tuberculosis (false positive), and another case of tuberculosis image being classified as normal (false negative). The exceedingly low number of errors made by the model further emphasized its capacity to retain high sensitivity and specificity despite the class imbalance present in the dataset.

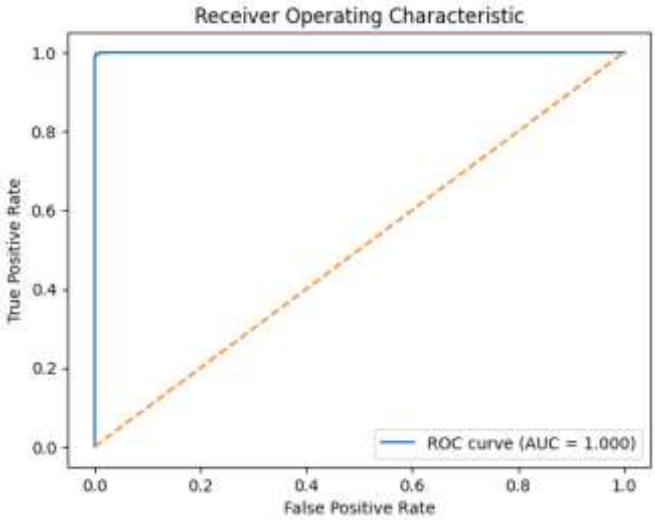

Fig 4.14 ROC curve of Swin Transformer

The ROC curve for this particular model is shown in Fig 4.14. This particular curve demonstrates the capability of the model in differentiating between the Normal and Tuberculosis cases as the threshold level changes. As can be seen from this curve, there is an almost perfect upward shift towards achieving a true positive rate of 1.0 and then a perfectly horizontal run. This implies that the model has perfect discrimination power in detecting TB cases. An Area Under the Curve (AUC) of 1.000 obtained indicates that the model has a perfect classification power since it perfectly ranks TB images over normal images at all threshold levels.

CAPTUM EXPLAINABILITY

Captum is a very useful open-source library for interpreting machine learning models provided by Facebook AI, targeting PyTorch-based models. It offers many attribute algorithms which enable to understand the importance of different input features for the output result produced by the model. In Tuberculosis detection tasks, Captum is particularly useful since understanding the reasoning behind a specific prediction made by a model is of great significance.

With Captum, attribute maps can be created to find out the regions of the chest X-ray image influencing the model's decision. By analyzing such attribute maps, it becomes possible to

make sure that the model actually pays attention to the correct lung regions and not to any irrelevant areas.

There are many attribute algorithms available in Captum, namely Integrated Gradients, Gradient SHAP, DeepLIFT, Layer Conductance, Saliency Maps, and others. The most commonly used one is Integrated Gradients because it produces smooth attributions and solves problems associated with gradient saturation.

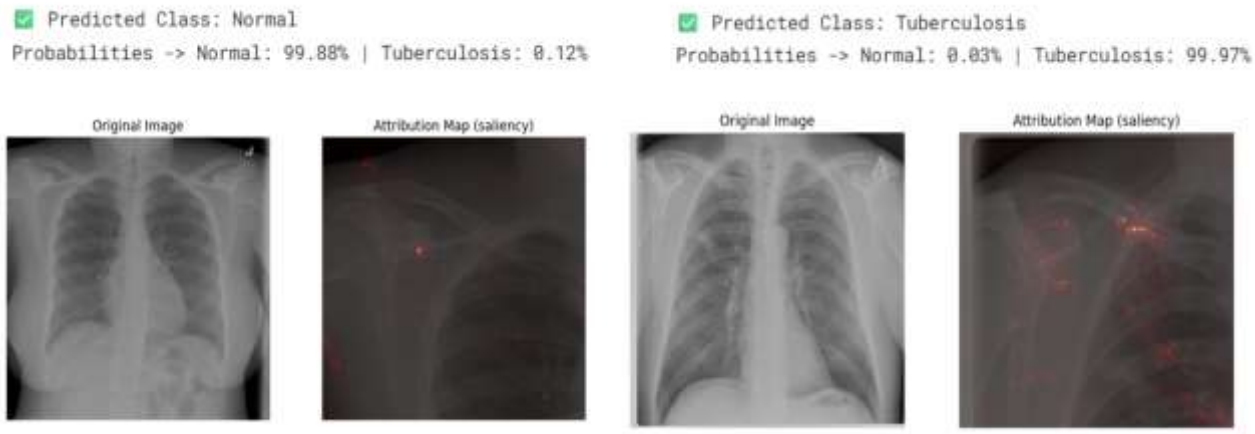

Fig 4.15 Captum-based attribution heatmaps for Swin Transformer model.

Fig 4.15 represents the attributions from the Swin Transformer model using Captum Explainability method. In the first column, there is the original chest X-ray that was predicted as a Normal case, and its attribute maps display few highlighted areas, implying the absence of any abnormal condition. The second column contains the original X-ray picture labeled as TB infection, and its attributed map demonstrates crucial lung areas colored in deep red hues.

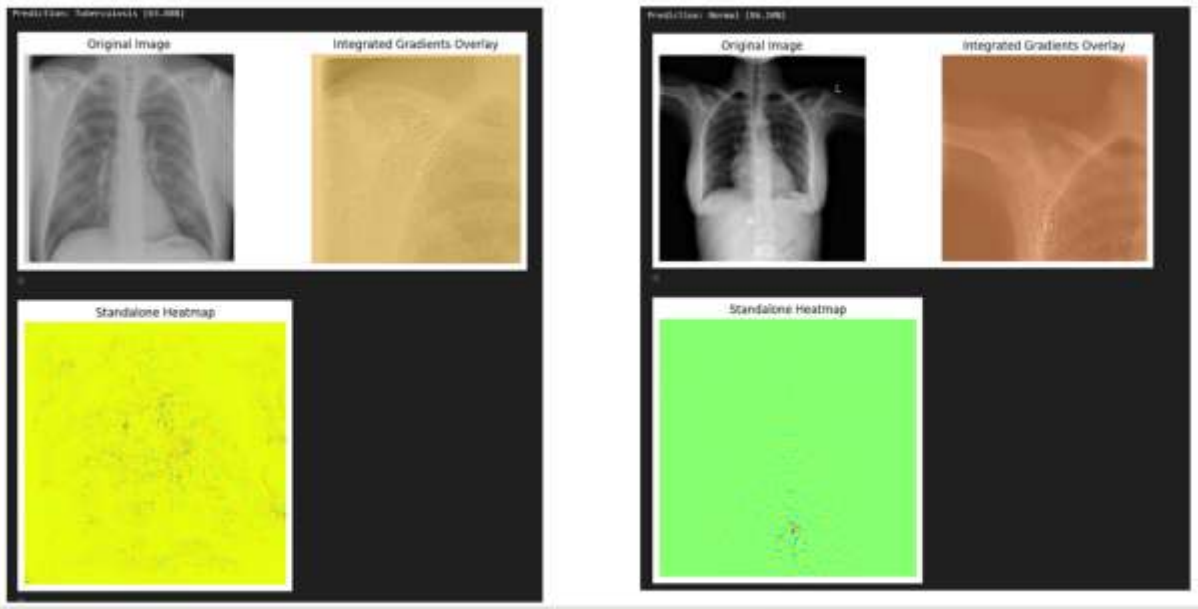

Fig 4.16 Captum-based attribution heatmaps for ConvNeXt model

This is seen in Fig 4.16 where there is an illustration of the outputs for both TB and Normal chest X-ray images produced from the Captum Integrated Gradients explainability approach for the ConvNeXt model. For the TB diagnosis, it is clear that there is activation in the upper and central lungs and, therefore, showing that there is pathology which includes infiltrate or opacities that are consistent with TB infections. As depicted in the overlay visualization, it becomes evident that there is clear indication of the specific lung regions that contribute significantly to the positive diagnosis. Conversely, in the normal case, the activation is less prominent and this means that there are no abnormalities in the lung regions hence, the correct diagnosis.

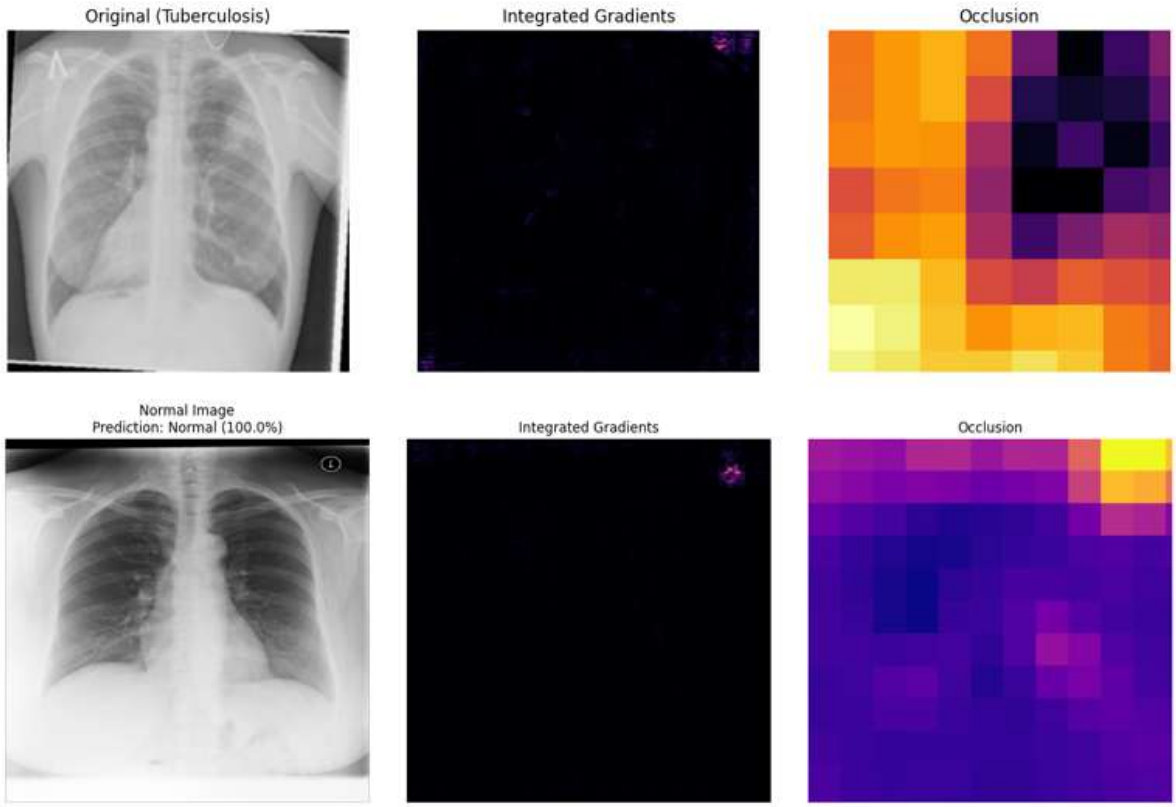

Fig 4.17 Captum-based attribution heatmaps for Vision Transformer model

These are the Fig 4.17 images that illustrate the vision transformer models for the explainability techniques of the integrated gradients and occlusion. In the case of tuberculosis, where there is a problem with the lungs (upper image), the occlusion map shows dark areas, while the integrated gradient shows less activation. However, in the case of normal health (lower image), there is no activation on both maps.

Fig 4.18 Captum-based attribution heatmaps for Ensemble model

Explanation obtained via Captum's Integrated Gradients technique for the image of the normal chest X-ray is shown in Fig 4.18 below. As can be seen from the figure, the areas with the highest contribution to the prediction made by the model are emphasized. It can be seen that the model mainly pays attention to the lungs, at the same time evenly distributing its attention around the chest region. In addition, there are no bright or abnormal areas of emphasis, implying that there are no abnormal patterns on the image. In other words, this explains the correct detection of the structure by the model without being affected by unnecessary information.

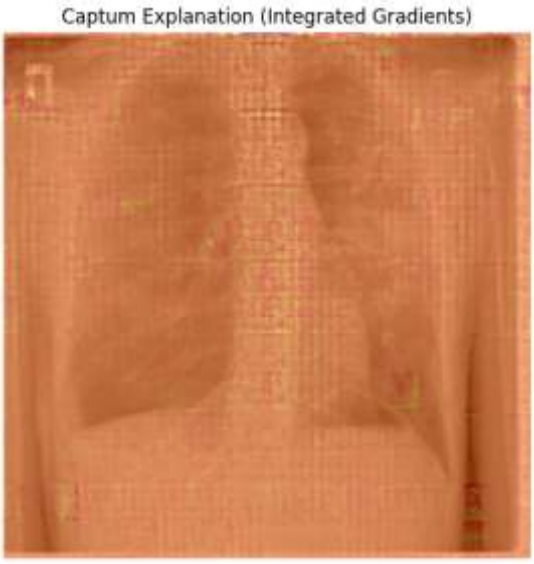

Fig 4.19 Captum-based attribution heatmaps for Ensemble model

Fig 4.19 illustrates the explanation obtained through the use of Captum's Integrated Gradients technique for a tuberculosis X-ray image predicted by the ensemble model consisting of ConvNeXt and Vision Transformer architectures. The visualized result shows the regions that play an essential role in the process of prediction. From Figure 4.19, it can be seen that the model is paying closer attention to particular regions of the lungs that contain abnormalities such as opacities and uneven textures, thus implying the presence of pathological changes in the patient. In contrast to the normal case, the attention regions are localized and focused, which demonstrates the model's ability to identify disease-specific changes.

GRADCAM++ EXPLAINABILITY

Grad-CAM++ (Gradient-weighted Class Activation Mapping++) is a state-of-the-art Explainable AI method that can be employed to produce visual explanations of the important image regions affecting the predictions made by a deep neural network. Compared to conventional Grad-CAM, Grad-CAM++ offers better localization of important regions for complex applications. In this project, Grad-CAM++ has been implemented for architectures like ConvNeXt, Vision Transformer and Swin Transformer to produce heat maps based on images of the chest X-ray dataset. Heat maps will emphasize the important image regions, especially those corresponding to lungs that are significant in classifying Normal and Tuberculosis patients.

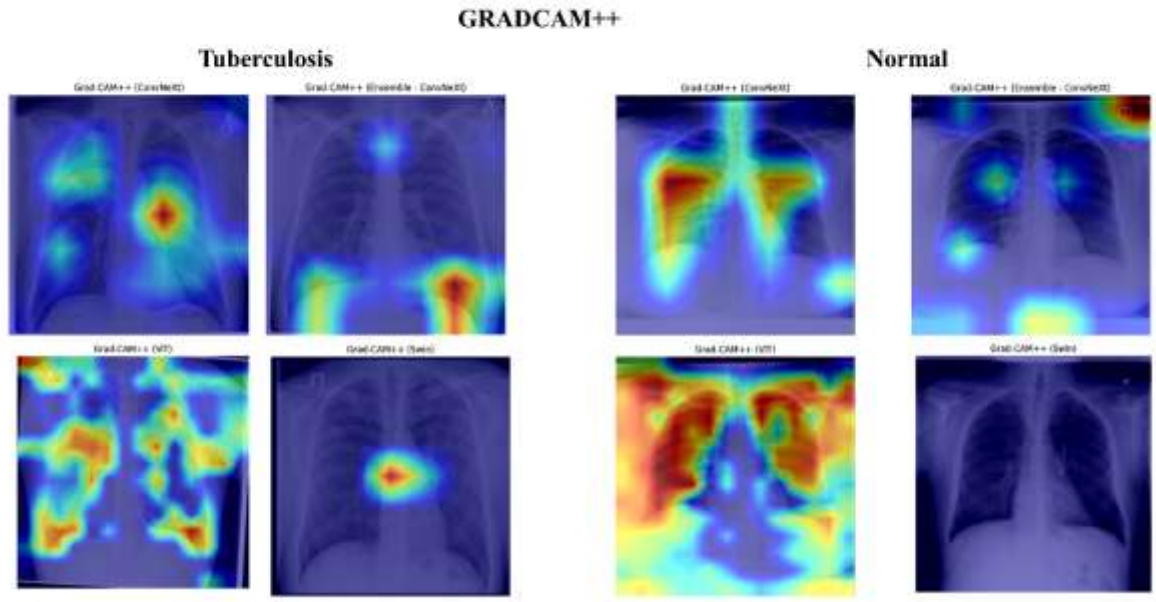

Fig 4.20 Gradcam++ Explainability for Tuberculosis and Normal Cases

Fig 4.20 shows the Grad-CAM++ visualization of both Tuberculosis and normal chest X-ray image for different models including ConvNeXt, Vision Transformer , Swin Transformer and

ensemble models. Heat maps demonstrate areas that have the most impact on model prediction. For Tuberculosis, all models localize their attentions in certain localized areas in the lung that represent abnormality, for instance opacities and lesions with concentrated activation. While for normal images, no hot spots can be seen, which demonstrates the absence of the disease. From all models, ensemble models perform better localization. All in all, based on the results shown above, Grad-CAM++ clearly states that all models pay attention to clinically meaningful regions.

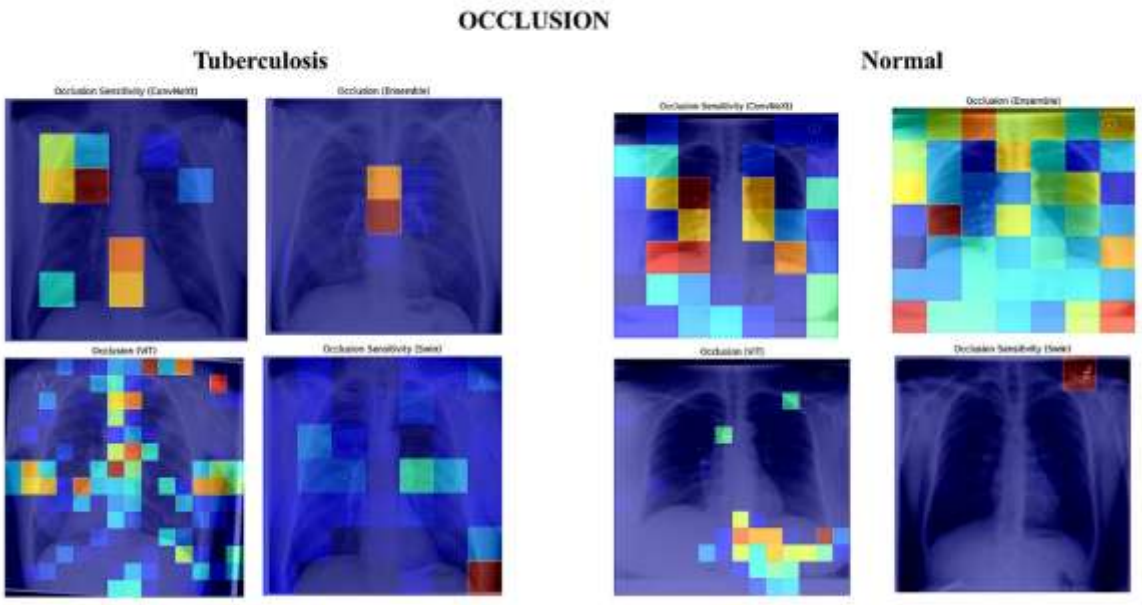

Fig 4.21 Occlusion Explainability for Tuberculosis and Normal Cases

The occlusion sensitivity map for both the Tuberculosis and Normal cases for various models including ConvNeXt, Vision Transformer, Swin Transformer, and the ensemble model is shown in Fig 4.21. The occlusion method is used to determine the key areas by masking specific portions of an image and determining how it affects the performance of a neural network to make predictions based on occluded portions of an image. For Tuberculosis images, there exists some significant sensitivity in the localized portions of the lung region because the occluded portion of the image will lead to a significant reduction in the confidence of making a correct prediction. However, for the normal images, sensitivity maps are much less pronounced, hence implying the absence of any irregularities. Among all the models considered in this study, the ensemble model gives the best results due to the inclusion of various model strengths.

Fig 4.22: User Interface of TB Detection Web Application

The TB Detection Web Application, which enables automatic tuberculosis detection by analyzing X-ray images of the patient's chest, has its graphical user interface presented in Fig 4.22. Specifically, the GUI of the proposed application allows the users to select the appropriate model and upload the X-ray image either via drag-and-drop or choosing from the list of files. Multiple deep learning models such as ConvNeXt, ViT, Swin Transformer, and an ensemble approach are implemented in order to improve the quality of analysis. Once the image is uploaded, the user will be able to start the prediction procedure by clicking on the 'Analyze Chest X-Ray' button. As previously mentioned, the suggested web application is supposed to act as an AI-driven clinical decision support system, meaning that accurate predictions will be accompanied by explainable AI features. Lastly, the system includes a clinical disclaimer that stresses that the application was not intended for diagnostic purposes but rather support ones.

Fig 4.23: Prediction Results Display of TB Detection Web Application

As shown in Fig 4.23, the output screen of the predictions performed by the TB Detection Web Application includes the diagnosis result details for a specific chest X-ray image. It includes the clinical diagnosis of Normal with a confidence score of 93.3%, suggesting that the accuracy of model prediction is high in this case. Moreover, it includes predictions performed by several models including ConvNeXt, Vision Transformer, Swin Transformer, and ensemble model. Here, the Swin Transformer gets the highest confidence score and is thus considered to be the best model. Also included in this interface is the agreement status of models, making it easy for users to understand the consistency among different models. Apart from that, the uploaded chest X-ray image is displayed along with the diagnosis results.

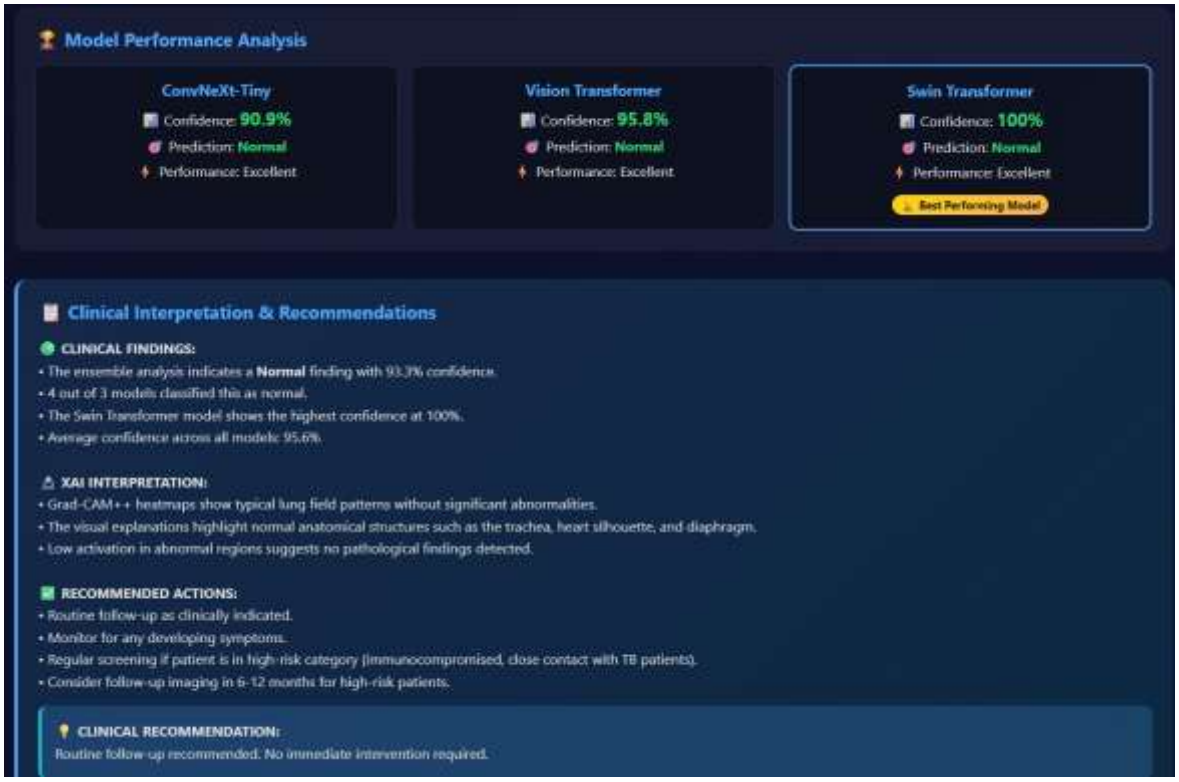

Fig 4.24: Model Performance Analysis and Clinical Recommendations Interface

The TB Detection Web Application model performance analysis and clinical interpretation module is illustrated in Fig 4.24 below. It gives a comparative analysis of different models such as ConvNeXt, Vision Transformer, and Swin Transformer. All of these models come up with their confidence levels and outputs and, among these, the Swin Transformer is noted to be the best performing with the highest confidence level. The summary of the clinical findings shows a normal condition with high consistency between all models together with high confidence levels. The XAI interpretation of the findings shows that Grad-CAM++ images concentrate only on normal anatomical regions without focusing on any abnormalities, supporting the prediction made. In addition, the interface indicates clinical actions recommended for the

patient, such as follow-up and monitoring, while the final clinical conclusion recommends that no action should be taken yet.

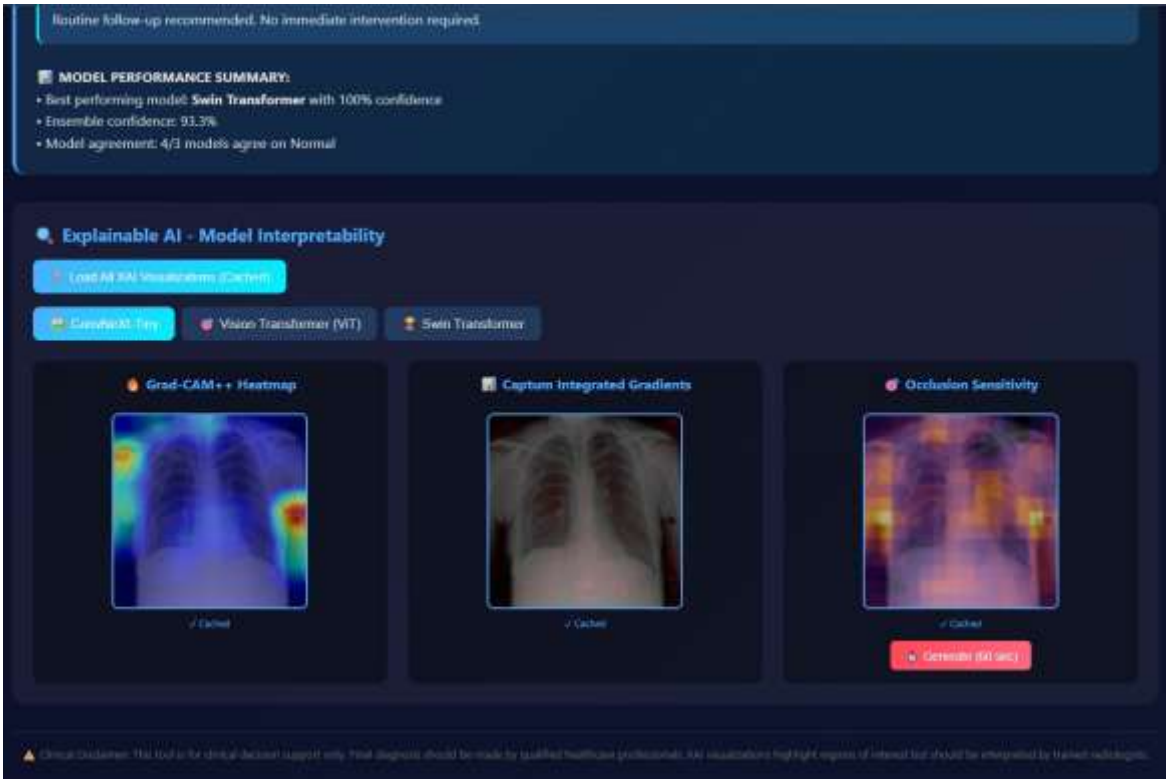

Fig 4.25: Explainable AI (XAI) Module for Model Interpretability

Fig 4.25 depicts the visual interface of Explainable AI of the TB Detection Web Application. This interface offers several methods for explaining the prediction made by the model, such as Grad-CAM++, Captum Integrated Gradients, and Occlusion Sensitivity, which have been applied on models like ConvNeXt, Vision Transformer , and Swin Transformer. These techniques help visualize which sections of the chest X-ray image played crucial roles in predicting the output from the trained model. Grad-CAM++ helps to show specific areas where attention was drawn while predicting the output, Captum helps show attribution per-pixel of the input, and occlusion sensitivity finds regions of high significance through masking different areas of the image.

4.3 RESULT ANALYSIS

Model Summary

Models considered for deep learning are ConvNeXt Tiny, Vision Transformer, Swin Transformer, and an ensemble of the three models. The ensemble model performed better than the other three models in detecting TB from the TB Chest X-Ray dataset.

Table 4.1 Result analysis for models

MODEL	TEST ACCURACY
ConvNeXt	**97.29%**
Ensembled	**98.00%**
Vision Transformer	**99.43%**
Swin Transfomer	**99.71%**

Test Accuracy results of the different models used to classify TB are presented in Table 4.1 below. The ConvNeXt model recorded an accuracy of 97.29%. However, the Vision Transformer model proved to be more accurate with 99.43%. The highest accuracy among the three models stood at 99.71% from the Swin Transformer, thus showing high capabilities of capturing both local and global lung characteristics. The best ensemble approach that combines ConvNeXt and Swin Transformer was recorded at 98%. Generally, the Swin Transformer surpasses other models as far as TB classification accuracy is concerned.

4.4 OBSERVATIONS FROM THE WORK

Based on the results obtained from the experiments conducted, it is apparent that deep learning models have outstanding capabilities in terms of diagnosing Tuberculosis (TB) from chest X-rays. This is because the models exhibited high accuracy in recognizing both low-level and high-level features for classifying the data into either diseased or healthy conditions. As for individual models, the best performing model was the Swin Transformer model, whose superiority could be attributed to its hierarchical features and attention mechanism. ViT and ConvNeXt followed immediately, whose success came due to their capabilities in capturing dependencies and local features respectively.

The performance of the models was even better when the ensemble model was used, which helped to increase accuracy by taking advantage of all the models at once. This made the models very stable and reliable as the ensemble model would help reduce the chances of making mistakes.

One interesting observation is that all models had consistent and stable convergence while training. From the graphs obtained for training and validation losses, there is no sign of overfitting and underfitting, which can be attributed to successful use of such techniques as

data augmentation, using weighted loss function, and early stopping. Such approach guarantees generalizability of the model.

It has been found from the confusion matrix results that there are very low occurrences of both false positives and false negatives, which means that the models are highly accurate in their prediction task. This property is especially important when considering application in the field of medical diagnosis. Additionally, high ROC curves, whose Area Under Curve (AUC) values are near one, further validate the strong ability of the models to discriminate TB patients from the healthy population.

Explainable AI techniques like Grad-CAM++ method, Captum (Integrated Gradients), and Occlusion sensitivity were used successfully to understand how the model makes decisions. It was discovered that all the heatmaps and attribution maps produced by XAI techniques indicate that the model's attention was concentrated only on the lung region, which is quite logical and medically reasonable area.

Moreover, it was seen that applying several XAI methods together provides complementing viewpoints, thus making it easier to have a more thorough insight into how a model behaves. In particular, while Grad-CAM++ identifies key regions, Integrated Gradients give information about which pixels contribute to a certain prediction, and Occlusion Sensitivity acts as a way to verify that different image parts are indeed relevant.

Regarding implementation, it should be noted that building such a web app proves that the proposed system can be practically used in real-life applications. By uploading chest X-rays and obtaining both predictions and explanations of these predictions in real time, the model becomes more user-friendly, thus acting as an assistive tool for medical professionals who want to make sure of their diagnosis.

All in all, the findings prove that integrating advanced models of deep learning, ensemble approaches, and explainable AI allows building a highly effective, robust, and, most importantly, interpretable system for TB detection.

4.5 SUMMARY

The obtained results from the study clearly show that the modern deep learning approaches can be highly accurate and reliable in TB diagnosis with the help of chest X-rays. Out of all the tested deep learning models, the Swin Transformer gave the best results, followed by ViT and ConvNeXt. With the help of ensembling, better stability and consistency in predictions have been obtained. The performance of the models through accuracy, precision, recall, and ROC-AUC metrics showed that all the tested models performed well without any errors in classification. Besides, the smooth convergence of all models was observed during training

because they had very low loss values, showing no overfitting in their performance. Further, the application of explainable AI techniques in the proposed methodology proved that the predictions made by the deep learning models were highly accurate, as they used clinical relevant regions in X-rays, especially the lungs.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

In the current research, an intelligent framework that utilizes artificial intelligence (AI) for TB detection using chest x-ray images has been developed. The main goal of the project was the development of a reliable, accurate, and effective automatic system that could aid in early diagnosis, thereby minimizing the requirement of human intervention in the process. For accomplishing the above goals, deep learning networks such as ConvNeXt-Tiny, Vision Transformer (ViT), and Swin Transformer were used

All of these methods were able to extract complex information from the input images effectively. While the ConvNeXt network managed to extract local spatial information, the other two algorithms relied on self-attention methods to capture global dependencies. In order to maximize efficiency and reliability, an ensemble technique was used.

It can be noted from the experimental results that all the developed models were found to perform very well based on the performance indicators, including accuracy, precision, recall, and F1-score. Out of all the models, Swin Transformer was found to have performed best owing to the effective hierarchical feature extraction process and attention mechanism. Ensemble Model though a bit expensive computationally was able to deliver better results in terms of accuracy and reliability.

An important innovation in the current study is the implementation of several explainable AI (XAI) methodologies such as Grad-CAM++ method, Captum (IG), and Occlusion Sensitivity analysis. The use of the above-mentioned XAI methods helped in generating visualizations that pointed out the important areas in the chest X-ray image, contributing to the model predictions. The heatmaps generated using these methods showed that the models paid attention to the lung areas mostly, which seems to be expected from the clinical point of view.

Apart from designing the models, a user interface in the form of a website was developed to demonstrate that the developed system can be easily used for practical purposes. The application allows to make predictions in real-time and visualize explanation maps and recommendations.

Generally, the results of this research project prove that the combination of deep learning approaches, ensemble learning methods, and explainable artificial intelligence enables the creation of an effective and efficient TB detection model. The framework introduced in this

paper has great potential to facilitate early detection of TB, especially in developing countries where specialists cannot be easily accessed.

5.2 FUTURE WORK

Despite the promising performance that the suggested system has shown, there remain several ways to improve and extend its capabilities. In the first place, one can work on expanding the used dataset by adding more diverse chest X-rays of varying origins and taken under various conditions at other institutions.

Another direction in which the system can be improved is related to optimizing the utilized deep learning architecture in order to decrease computational requirements and increase the speed of predictions. Various approaches including model pruning, quantization, and knowledge distillation can be used to optimize the utilized deep learning models.

Finally, there is also room for extending the ensemble method by using other state-of-the-art network architectures in addition to more sophisticated techniques of feature fusing.

Explainable AI is another avenue for future research that can yield valuable results. While existing methods allow for a good visualization, more comprehensive and clinically meaningful explanations can be constructed to aid decision-making in medicine. The combination of several XAI methods within one unified approach will enable the researcher to acquire a deeper understanding of how the model works.

In terms of scope, the system can be generalized to detect not only Tuberculosis but other pulmonary diseases as well, such as pneumonia, lung cancer, and even COVID-19. Thus, a multi-disease diagnosis system will be designed.

Regarding deployment, the system can be modified from the existing web app prototype to the full-fledged CDSS. Features such as managing records, generating reports, integration with hospital databases, and data security should be included in the final product. Additionally, the system can be deployed in the cloud and on edge computing devices.

Continuous learning by means of real-time feedback provided by healthcare professionals is another technique that could be used to enhance the functioning of the system. Introduction of active learning components would help to adjust the model to new data and scenarios.

Lastly, for practical applications, validation and certification of the system are crucial. Validation and certification with the involvement of healthcare professionals are needed in future research.

While there have been many positive outcomes obtained using the proposed framework, there are several weaknesses that should be noted. The first one is related to the performance of the framework based on the dataset used, and the size and diversity of the dataset may not cover

all possible clinical cases. Another weakness of the proposed model is the fact that the system uses only images taken with chest x-rays, without including any information regarding the patient's history or other clinical factors. Still, the proposed methodology can be considered a basis for developing smart diagnosing tools.

REFERENCES

[1] Özkurt, C. (2024). Improving tuberculosis diagnosis using explainable artificial intelligence in medical imaging. *Journal of Mathematical Sciences and Modelling*, 7(1), 33–44.

[2] Ali, R. W., Khatoon, S., Shazadi, S., Abbas, S., & Nazir, Z. (2026). Multi-source chest X-ray datasets for accurate and explainable tuberculosis (TB) diagnosis. *Spectrum of Engineering Sciences*, 4(3), 1574–1592.

[3] Acharya, V., Dhiman, G., Prakasha, K., Bahadur, P., Choraria, A., M, S., et al. (2022). AI-assisted tuberculosis detection and classification from chest X-rays using a deep learning normalization-free network model. *Computational Intelligence and Neuroscience*, 2022, 2399428.

[4] Agumba, J., Erick, S., Pembere, A., & Nyongesa, J. (2026). A deployable explainable deep learning system for tuberculosis detection from chest X-rays in resource-constrained high-burden settings. *medRxiv*, 2026.

[5] Vanitha, K., Mahesh, T. R., Kumar, V. V., & Guluwadi, S. (2025). Enhanced tuberculosis detection using Vision Transformers and explainable AI with a Grad-CAM approach on chest X-rays. *BMC Medical Imaging*, 25(1), 96.

[6] Rafi, M. S. M., Ali, M. S., Hossen, M. M., Sneha, S. Z. R., & Arefin, M. S. (2025). Chest X-ray analysis for pneumonia and tuberculosis detection using CNNs and Vision Transformers with explainable AI. In *2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)* (pp. 1–6). IEEE.

[7] Maheswari, B. U., Sam, D., Mittal, N., Sharma, A., Kaur, S., Askar, S. S., & Abouhawwash, M. (2024). Explainable deep-neural-network supported scheme for tuberculosis detection from chest radiographs. *BMC Medical Imaging*, 24(1), 32.

[8] Nansamba, B., Nakatumba-Nabende, J., Katumba, A., & Kateete, D. P. (2025). A systematic review on application of multimodal learning and explainable AI in tuberculosis detection. *IEEE Access*.

[9] Nafisah, S. I., & Muhammad, G. (2024). Tuberculosis detection in chest radiograph using convolutional neural network architecture and explainable artificial intelligence. *Neural Computing and Applications*, 36(1), 111–131.

[10] Ayalew, A. M., Asnake, N. W., Demil, G., & Oussalah, M. (2025). An explainable hybrid deep learning system for tuberculosis detection with Grad-CAM. *Discover Computing*, 28(1), 262.

[11] A. Wong, J. R. H. Lee, H. Rahmat-Khah, A. Sabri, A. Alaref, and H. Liu, "TB-Net: A tailored, self-attention deep convolutional neural network design for detection of tuberculosis cases from chest X-ray images," *Frontiers in Artificial Intelligence*, vol. 5, p. 827299, 2022.

[12] G. Siddharth, A. Ambekar, and N. Jayakumar, "Enhanced CoAtNet-based hybrid deep learning architecture for automated tuberculosis detection in human chest X-rays," *BMC Medical Imaging*, vol. 25, no. 1, p. 379, 2025.

[13] C. R. Dhivyaa, K. Nithya, C. S. Kumar, and R. Sudhakar, "Explainable model of fusion network with enhanced optimization approach for tuberculosis diagnosis," *IEEE Access*, vol. 12, pp. 176920–176937, 2024.

[14] Z. Z. Qin *et al*., "Tuberculosis detection from chest X-rays for triaging in a high tuberculosis-burden setting: An evaluation of five artificial intelligence algorithms," *The Lancet Digital Health*, vol. 3, no. 9, pp. e543–e554, 2021.

[15] M. Bhandari, T. B. Shahi, B. Siku, and A. Neupane, "Explanatory classification of CXR images into COVID-19, Pneumonia and Tuberculosis using deep learning and XAI," *Computers in Biology and Medicine*, vol. 150, p. 106156, 2022.

[16] Vanitha, K., Mahesh, T. R., Kumar, V. V., & Guluwadi, S. (2025). Enhanced tuberculosis detection using Vision Transformers and explainable AI with a Grad-CAM approach on chest X-rays. *BMC Medical Imaging*, *25*(1), 96.

[17] Ayalew, A. M., Asnake, N. W., Demil, G., & Oussalah, M. (2025). An explainable hybrid deep learning system for tuberculosis detection with Grad-CAM. *Discover Computing*, *28*(1), 262.

[18] Patel, M., & Badla, M. (2025). Chest X-ray-based tuberculosis detection using deep learning: An in-depth review of current models and future directions. *International Journal of Creative Research Thoughts (IJCRT), 13*(6), k607–k622

[19] Maheswari, B. U., Sam, D., Mittal, N., Sharma, A., Kaur, S., Askar, S. S., & Abouhawwash, M. (2024). Explainable deep-neural-network supported scheme for tuberculosis detection from chest radiographs. *BMC Medical Imaging*, *24*(1), 32.

[20] prasad Koyyada, S., & Singh, T. P. (2023). An explainable artificial intelligence model for identifying local indicators and detecting lung disease from chest X-ray images. *Healthcare Analytics*, *4*, 100206.

APPENDIX

Some screenshots of the Code Snippet:

```
# 🧠 6. Load ConvNeXt-Tiny and modify classifier
model = models.convnext_tiny(pretrained=True)
for param in model.features.parameters():
    param.requires_grad = False  # Freeze backbone

# Modify classifier
model.classifier[2] = nn.Linear(model.classifier[2].in_features, 2)
model = model.to(device)
```

Fig A1: Code snippet of Convnext Model

```
# Load pretrained ViT
vit_model = vit_b_16(weights=ViT_B_16_Weights.DEFAULT)

# Modify classifier for binary classification
in_features = vit_model.heads[0].in_features
vit_model.heads.head = nn.Linear(in_features, 2)
vit_model = vit_model.to(device)
```

Fig A2: Code snippet of Vit Model

```python
import torch
import torch.nn as nn

class EnsembleModel(nn.Module):
    def __init__(self, convnext_model, vit_model):
        super(EnsembleModel, self).__init__()
        self.convnext = convnext_model
        self.vit = vit_model

    def forward(self, x):
        out1 = torch.softmax(self.convnext(x), dim=1)
        out2 = torch.softmax(self.vit(x), dim=1)

        return (out1 + out2) / 2
```

Fig A3: Code snippet of Ensemble Model

```python
# ==========================
# 3) Build Swin Transformer
# ==========================
model = timm.create_model(CFG.model_name, pretrained=True)
# Reset classifier to 2 classes
model.reset_classifier(num_classes=2)
model = model.to(CFG.device)
```

Fig A4: Code snippet of Swin Transformer Model

16% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- 233 Not Cited or Quoted 16%
  Matches with neither in-text citation nor quotation marks
- 1 Missing Quotations 0%
  Matches that are still very similar to source material
- 6 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

11% Internet sources
11% Publications
0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.